

Turkish Aeronautical Association University



Term Project Report, Phase B Study (PDR)

[MADEM]

AST 454 Spacecraft Design - II



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Term Project Report, Phase B Study (PDR), [Project Acronym]

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3 Introduction

From the beginning of the Cold War, known as the Space Race, humans began to search for life resources in the heavens above. With the discovery of new planets, the question “Is there life in space?” gained much more attention. Nowadays, almost every space scientist has one dream. Going to Mars, known as the red planet. Since Mars has an atmosphere and recent research has proven that there once was water on Mars, it gained much more attention and importance as a livable planet.

The MADEM mission to Mars begins on the stationary Martian orbit which should be functional for a relay of at least 60 Earth days. Lander will be entering the atmosphere at 118.3 km Martian circular orbit and will land on the Apollinaris Mons Volcano. Lander will conduct experiments, analysis, and will take pictures of the Martian environment from one of the most important parts of the planet, a volcano.



Figure 1:Mons Apollinaris[t1]

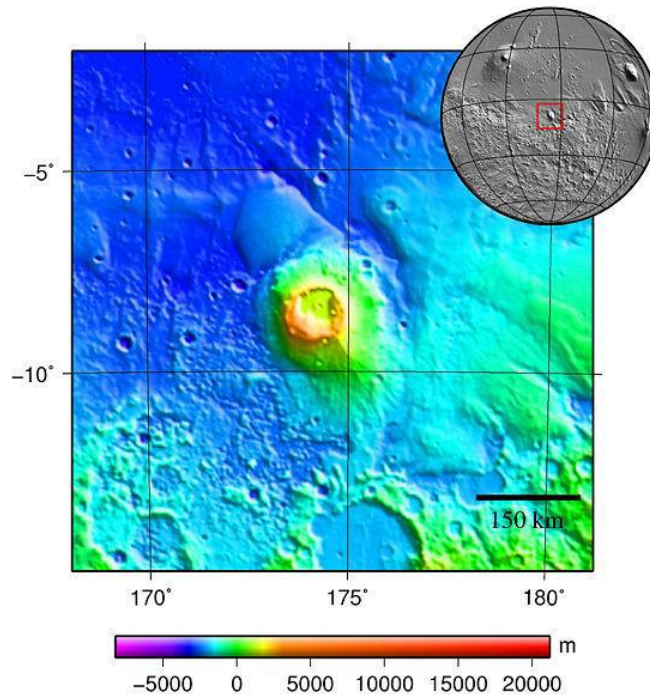


Figure 2: Topography of Mons Apollinaris [T2]

Mons Apollinaris Discovery and Exploration Mission (MADEM) is a Mars Robotic Lander designed to operate for two Earth months on volcano Mons Apollinaris. Apollinaris Mons is an ancient shield volcano in the southern hemisphere of Mars. It is situated near the equator. The volcano's caldera is named Apollinaris Patera.

Apollinaris Mons is about 5 kilometers high with its maximum elevation and it has a base about 296 kilometers in diameter. A caldera, which is around 80 km in diameter is at the top of this mountain. The volcano is approximately 3 billion to 3.5 billion years old. There is no artificial entry in Mars history after it ended up on a Volcano. [T3]

3.1 Name and Acronym of the Project

The Mons Apollinaris Discover and Exploration Mission (MADEM)

3.2 Team and Responsibilities

Doruk Demirci: Communications and Data-handling, Thermal Control

Kadir Hasan Taşkara: Mission Design, Mission Analysis, Entry, Descend and Landing (EDL)

Feyza Torun: Structural Design, Thermal Control

Tuğcan Baş: Thermal Control

Bayram Savlı: Payloads, Structural Design

3.3 The Mission Statement

At MADEM, our mission is to propel our capsule, launched by the Bus, through the journey to Mars, maneuvering its descent through a combination of parachutes and thrusters upon entering the Martian atmosphere. After a successful landing, our spacecraft will remain on the Red Planet for two months, meticulously collecting material samples and valuable data.

Our primary objective is to strengthen space exploration endeavors, pioneering a path toward a brighter future. By conducting this mission, we aim to contribute significantly to the advancement of space science and technology, paving the way for groundbreaking achievements. Each moment spent on Mars not only enhances our understanding of the cosmos but also serves as a crucial step toward establishing a robust foundation for future endeavors.

Through continuous efforts and the acquisition of knowledge, we aspire to bolster our expertise in space exploration, fostering innovation and pioneering new frontiers. Our commitment is not just to the present mission but to the enduring impact it will have on the progression of humanity's presence in space, allowing us to confidently stride into a future filled with limitless possibilities. [T4]

3.4 Background Information

[1]During our mission, we will conduct research on Mars' magnetic field, study its magnetosphere and ionosphere, investigate tectonic activities, analyze impact events, explore the subsurface, collect samples, and capture surface images. During our mission, we will conduct research on Mars' magnetic field, study its magnetosphere and ionosphere, investigate tectonic activities, analyze impact events, explore the subsurface, collect samples, and capture surface images.

Mars' magnetic field, characterized as weak and patchy, will be scrutinized using a magnetometer during our mission. This instrument aids in comprehending the planet's magnetic properties, history, and the dynamic interplay between the Martian magnetosphere and solar wind.

The magnetometer extends its utility to magnetospheric and ionospheric studies, elucidating Mars' interaction with solar wind and its atmospheric implications. Additionally, examining mars quakes provides insights into the planet's tectonic activity, pivotal for unraveling its geological history and determining the presence of Earth-like plate tectonics on Mars.

Fig. 19 The SEIS EBox. A ~ 5 kg and ~ 9 W electronic box. The E-box is 243.8 mm in height. The top is 303.5 mm × 125.5 mm while the bottom is 343.5 mm × 169.5 mm due to mounting structures



Figure 3:SEIS Box of Insight[T5]

There will be a robotic arm equipped with essential tools, including drills, scoops, and cameras, to facilitate the meticulous collection of Martian soil, rocks, or regolith. These gathered samples play a pivotal role in scientific analysis, enabling the exploration and investigation of potential indicators of past or present life on Mars. [T6]



Figure 4:Robotic Arm used in Insight[T5]

The camera, instrumental in landing site assessment, offers real-time imagery aiding mission controllers and scientists in evaluating safety and suitability. Additionally, during surface imaging, the camera captures unique perspectives, offering detailed views of the landing site. These images assist in hazard identification, geological evaluation, and topographical analysis for a comprehensive understanding.

3.5 Similar Missions

Table 1: Similar missions

Mission	Date of Launch	Similarity	Notes
Beagle 2	2 June 2003	Mass Budget	Failed
MER Spirit	10 June 2003	Landing Site	Operated 2208 sols
Schiaparelli EDM	14 March 2016	Design	Failed

3.5.1.1 Beagle 2

The Beagle 2 is an inoperative British Mars lander that was transported by the European Space Agency's 2003 Mars Express mission. It was intended to conduct an astrobiology mission that would have looked for evidence of past life on Mars.

The spacecraft was successfully deployed from the Mars Express on 19 December 2003 and was scheduled to land on the surface of Mars on 25 December. ESA, however, received no communication from the lander at its expected landing time on Mars, and declared the mission lost in February 2004 after numerous attempts to contact the spacecraft were made.

The Beagle 2's fate remained a mystery until January 2015, when it was located on the surface of Mars in a series of images from NASA's Mars Reconnaissance Orbiter HiRISE camera. The images showed it landed safely but two of its four solar panels failed to deploy, blocking the spacecraft's communications antenna.

The Beagle 2 is named after HMS Beagle, the ship that took the naturalist Charles Darwin on his round-the-world voyage. [T8]

3.5.1.2 MER Spirit

NASA's Mars Exploration Rover (MER) mission was a robotic space mission involving two Mars rovers, Spirit and Opportunity, exploring the planet Mars. It began in 2003 with the launch of the two rovers to explore the Martian surface and geology; both landed on Mars at separate locations in January 2004. Both rovers far outlived their planned missions of 90 Martian solar days: MER-A Spirit was active until March 22, 2010, while MER-B Opportunity was active until June 10, 2018. [T9]

3.5.1.3 Schiaparelli EDM

Schiaparelli EDM was a failed Entry, Descent, and Landing Demonstrator Module (EDM) of the ExoMars programme a joint mission of the European Space Agency (ESA) and the Russian Space Agency Roscosmos. It was built in Italy and was intended to test technology for future soft landings on the surface of Mars. It also had a limited but focused science payload that would have measured atmospheric electricity on Mars and local meteorological conditions. Launched together with the ExoMars Trace Gas Orbiter (TGO) on 14 March 2016, Schiaparelli attempted a landing on 19 October 2016. Telemetry signals from Schiaparelli, monitored in real time by the Giant Metrewave Radio Telescope in India (and confirmed by Mars Express), were lost about one minute from the surface during the final landing stages. On 21 October 2016, NASA released an image by the Mars Reconnaissance Orbiter showing the lander's crash site. The telemetry data accumulated and relayed by ESA's ExoMars Trace Gas Orbiter and Mars Express were used to investigate the failure modes of the landing technology employed. [T10]

4 Plans

For the sake of completeness, the management plan, system engineering plan, and product assurance plan are mentioned in the appendix.

5 Technical Requirements Specification

[The technical requirements finalized in SRR should be here. Classify the requirements under headings like general, user, performance, operational requirements etc. and give codes to requirements.]

General Requirements

- GEN-01 MADEM lander shall land on Mars.
- GEN-02 MADEM lander shall take panchromatic images of Mars.
- GEN-03 MADEM lander shall measure seismic activities and changes in Mars's magnetic field.
- GEN-04 MADEM lander shall collect and analyze Mars regolith. [08]
- GEN-05 An orbiter shall be released at Mars' areostationary orbit and orbiter shall function as a relay to Earth.

Performance Requirements

- PER-01 The seismometer shall detect seismic activities of Mars up to 1e-9 meters.
- PER-02 The panchromatic camera shall take medium resolution panchromatic images of Mars surface.
- PER-03 The robotic arm shall collect regolith to analyze thoroughly.

- PER-04 The magnetometer shall sense changes in Mars' magnetic field.
- PER-05 The COM subsystem shall maintain a link margin of minimum 3 dB (for both downlink and uplink).
- PER-06 The EPS shall provide an average power of 396.286 W-h during sunlight.
- PER-07 The EPS shall provide an average power of 516.16 W-h during eclipse.
- PER-08 The EPS shall provide loads with regulated power at 28.6 V.
- PER-09 The EPS shall not exceed a DoD (Depth of Discharge) of 20% of battery.

Operational Requirements

- OP-01 MADEM lander shall operate for a minimum of two Earth months (60 days).
- OP-02 MADEM lander shall tolerate temperature swings from 20 to –153 degrees.
- OP-03 MADEM lander shall tolerate dust storms that can reach 100 km/h.

Constraints

- CON-01 MADEM lander shall weigh 100 kg (without margin) on the surface of mars.
- CON-02 Communications shall abide by ITU and FCC regulations.

6 Concept of Operations

As of 2024, all methods of landing on Mars have used an aeroshell and parachute sequence for Mars' atmospheric entry and descent. Although all entry and descent required aeroshell and parachute, landing sequences have differed. In this manner, there have been three different concepts of landing; the use of retrorocket, airbag, and sky crane.

Retrorockets are widely used in landers, not only in Mars landers but also in Earth's re-entry vehicles and Moon landers.

Airbags, however, are used when soft landing is required, but retrorockets cannot be used on the lander/rover due to their physical constraints. In such a case, airbags absorb the shock until the spacecraft comes to rest.

Sky cranes on the other hand are used in Mars' landing history just twice, by Curiosity and Perseverance rovers of NASA. Sky crane is a platform with eight engines that lowers the rover on three nylon tethers until the soft landing.

Figures listed below show each type of concept in detail. [K1N]



Figure 5:Retrorockets, airbags and sky crane, respectively[K2N]

6.1 Mission Timeline

The Martian year comprises 668.59 Martian days, also called sols, which is equivalent to 687 Earth days. Each sol is 24 hours and 37 minutes in length; 37 minutes longer than a day on Earth. The rotational axes of Mars and the Earth are tilted at similar angles about their orbits' planes. This angle is known as the axial tilt, and it is 24.9° for Mars and 23.44° for the Earth. [K13]

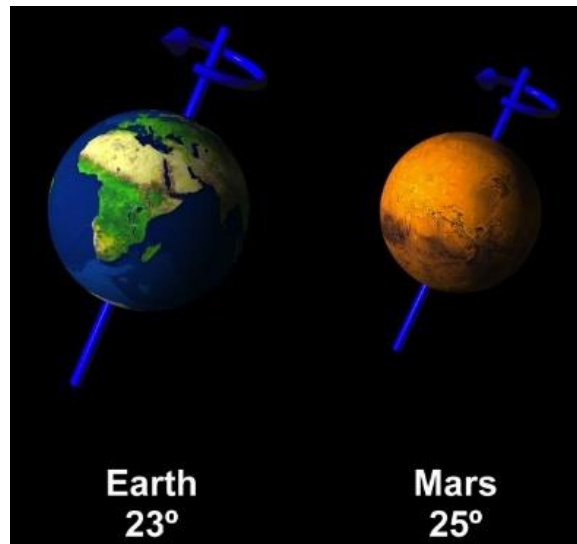


Figure 6: Illustration of the rotational axes of Earth and Mars[K14]

The seasons are of course reversed for the Southern Hemisphere just like Earth. Scientists mark the time of Mars year to use solar longitude, shown as L_s . The vernal equinox of Mars occurs when L_s is 0° , where the beginning of northern spring comes up. [K15]

Areocentric longitude (L_s)	Season	
	Northern Hemisphere	Southern Hemisphere
0° to 89°	Spring	Autumn (Fall)
90° to 179°	Summer	Winter
180° to 269°	Autumn (Fall)	Spring
270° to 359°	Winter	Summer

Table 2:Areocentric Longitudes and corresponding seasons[K16]

However, the eccentricity of Mars' orbit is larger than the Earth's, which causes the solar input to be significantly different when one pole is in sunlight than the other pole. Therefore, the summer in the southern hemisphere is much warmer than summer in the northern hemisphere because Mars is farther from the Sun during northern summer. [K17]

The most efficient time interval for MADEM is estimated to be between the solar longitudes of 251° and 270°, where the Martian perihelion starts and extends until the southern hemisphere summertime. Referred to Earth time, 08.05.2024-26.07.2024 are the optimum dates where the operation can occur, counting for at least 79 days and meeting the 60 Earth days requirement. For the astrodynamics estimations, Mars Climate Database has been highly used. [K20]

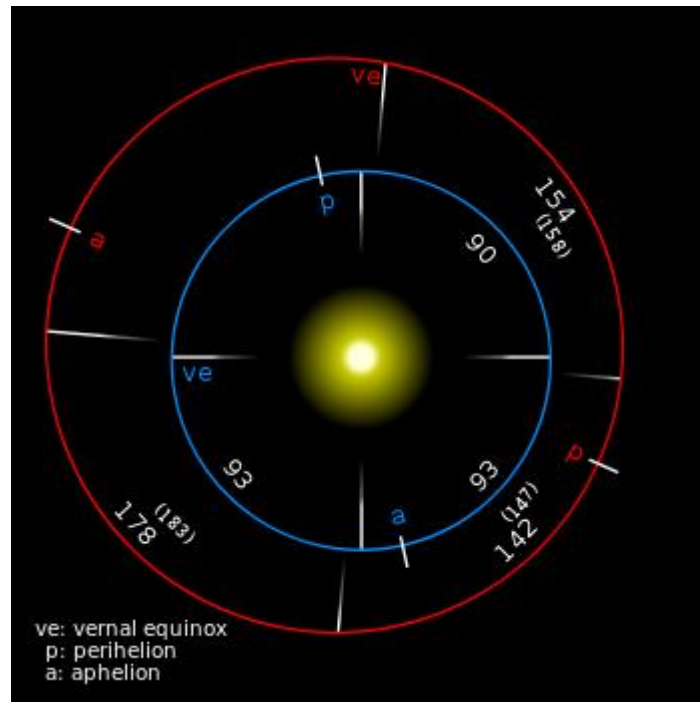


Figure 7: Martian season lengths and time as compared to seasons on Earth (in blue), with marks for the vernal equinox, perihelion, and aphelion [K18]

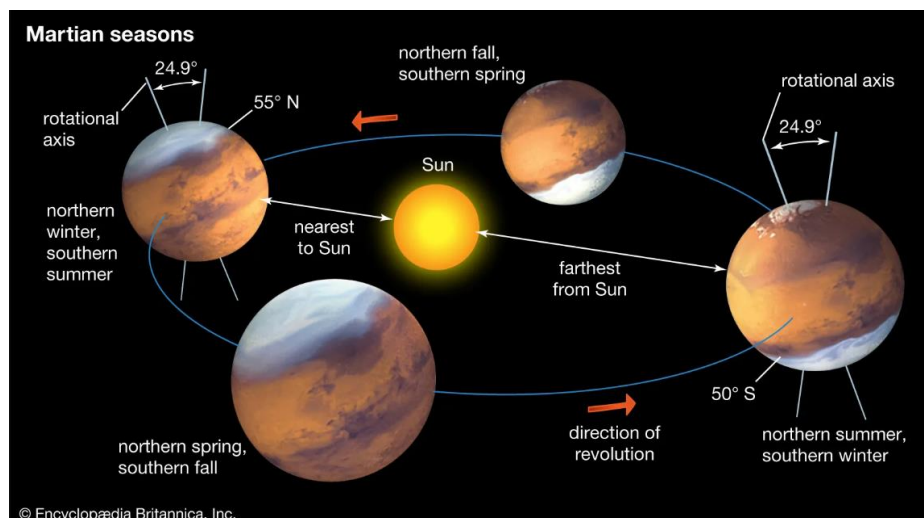


Figure 7: Seasons of Mars illustrated in orbit [K19]

A solar conjunction does not occur in 2024, therefore any signal loss is not included in MADEM. [K21]

6.2 Atmosphere

The atmosphere of Mars consists of CO₂ (95%), but other types of gases such as N₂, Ar and O₂. Its atmosphere is much thinner and colder than Earth's, having a max density 0.02 kg/m³ (about 2% of Earth's value) with a temperature generally down to 213K. The average surface pressure is about 610 pascals, which is less than 1% of the Earth's value.

The atmosphere's specific heat ratio is 1.29 and specific gas constant is 188.42 J/KgK. To compare, Earth's air has a specific heat ratio of 1.4 and its specific gas constant is 287.1 J/KgK. These lower properties of the atmosphere lead to slower speed of sound and thus higher Mach number when compared to Earth, which directly affects the parachute deployment sequence. [K3N]

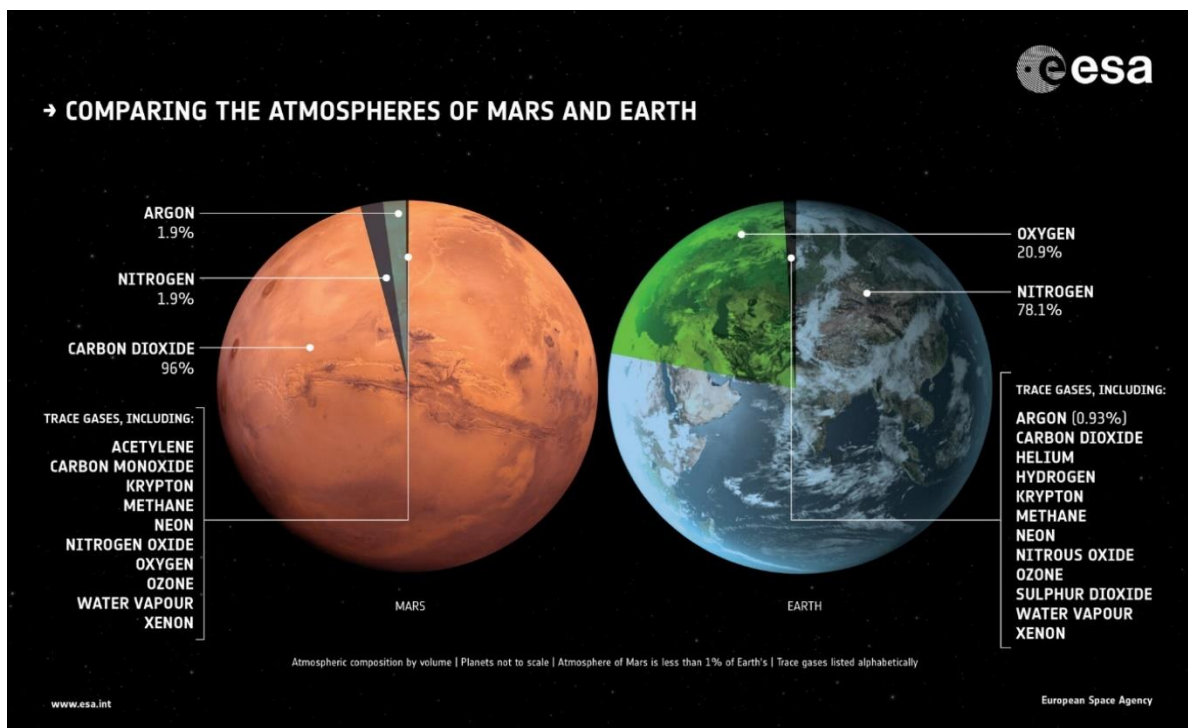


Figure 8: Mars and Earth Atmospheres [k4n]

In this project, a mathematical model of the atmosphere made by the Mars Global Surveyor is used. The values are given in metric units, where temperature is Celsius, and pressure is kilopascals. Density, of course, is found by equation of state. Figure 6.7 shows the atmosphere model in metric units. Note that this model is exponential.

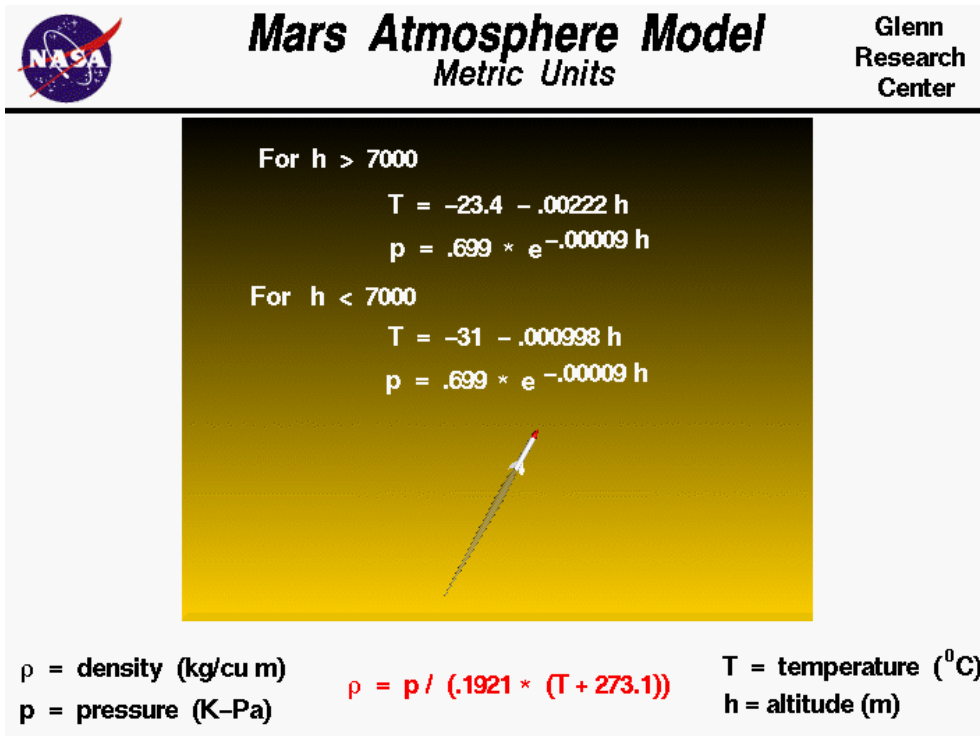


Figure 9: Mars Atmosphere Model [K5N]

7 The Chosen Mission Concept

7.1 Mission Trade-off

Sky crane is modern but very heavy and expensive concept compared to retrorockets and airbags. Although it satisfies sensitive landing sequence, this technique is not selected in trade-off.

In this section, the concepts of retrorockets and airbags are called concept 1 and concept 2, respectively. Taking the landing site into consideration, a mountain is an elevated portion of the ground no matter how smooth and plain caldera it has on the top of it. Such a formed requires accurate guidance, navigation, and control for right spot landing.

In the case of airbag usage, GNC is not much applicable compared to retrorockets. The bags limit all the equipment needed, such as a camera or radar altimeter. Therefore, selecting concept 2 will lead to a precise ballistic entry determination, which is not feasible even though it is achieved, because of the undetermined land shapes of the caldera and unknown number of bounces.

As a result, the first concept is selected due to guidance, navigation, and control concerns. It will also provide footage recording during descend and landing phase due to less constraints. Note that thanks to visual based navigation, recording and navigating will be possible at the same time.

Table 3: Trade-Off for the Concepts

 TRADE - OFF 	
 Concept 1	 Concept 2
It is less complex in terms of maintenance since it involves fewer moving parts. The maneuverability offered by it makes the landing more controllable, providing an advantage in dealing with unexpected situations.	The material surrounding it may introduce additional maintenance requirements and potential points of failure. The landing part of Concept 2 is maneuverability close to 0, making it very challenging to land in a desired location.

7.2 Concept 1

MADDEM will use closed-loop guidance, navigation and control system based on radar altimeter and visual based navigation, meaning that it will continuously regulate the system to maintain the desired state. It will be using shallow throttling thrust vector control, meaning there will not be a gimbaled thruster and the control is provided by reducing thrust from some of the engines, which increases reliability and decreases complexity.

7.3 Mission Analysis

The ΔV budget required for low Martian Orbit to Martian surface implies just one de-orbit burn, and the rest is satisfied by aerobraking. As estimated, the last speed of the spacecraft is supposed to be 0 since it ends up with landing. In mission analysis, it is supposed to determine the Martian low orbit velocity as ΔV budget is wanted to be most efficient. Assuming the orbit to be circular just before de-orbit burn, ΔV can be expressed just as the difference between the circular orbit velocity and landing velocity, which is zero. In this analysis, Mars equatorial radius is taken as 3397 km.

With appropriate interpolations, the convenient entry altitude is determined to be 118.3 km from the sea level. Unlike MADDEM, most of the Mars entry vehicles did not enter the atmosphere from the Martian orbit. They had gotten into Martian Transfer Orbit and just followed the interplanetary trajectory and applied retrograde burn to slow down.

$$V = \sqrt{\left(\frac{GM}{r}\right)}$$

The equation shown above gives the orbital speed for circular orbits, where G is the gravitational constant and equal to $6.67 \times 10^{-11} \text{ N m}^2 / \text{kg}^2$, M is the mass of the planet and r is the distance from the center of the planet to the orbiting object.

Mars has a mass of $6.39 \times 10^{23} \text{ kg}$ and an equatorial radius of $3.397 \times 10^6 \text{ m}$. [K9] From those values we obtain 3482 m/s or 3.4820 km/s for landing. MATLAB is used for this calculation and the codes are shared in Appendix A.

The reason 118.3 km of altitude is going to be shared in the following sections.

7.3.1 Entry Conditions

The southern portion of the Apollinaris Mons is higher and sharper compared to the northern portion, as shown in figure 2. Approaching the target point from a more elevated part might cause a hard crash in case the spacecraft misses the dedicated trajectory. Therefore, approaching from the northern side would bring higher reliability.

The orbit design also possesses a significant role. A near circular orbit is always more beneficial for entry sequence since its orbital velocity is lower than the eccentric orbit's velocity at the same periapsis point. Having a circular polar orbit will eventually bring shorter orbital periods and global coverage.

Therefore, the entry vehicle will be entering the atmosphere from Mars's circular polar orbit's northern part.

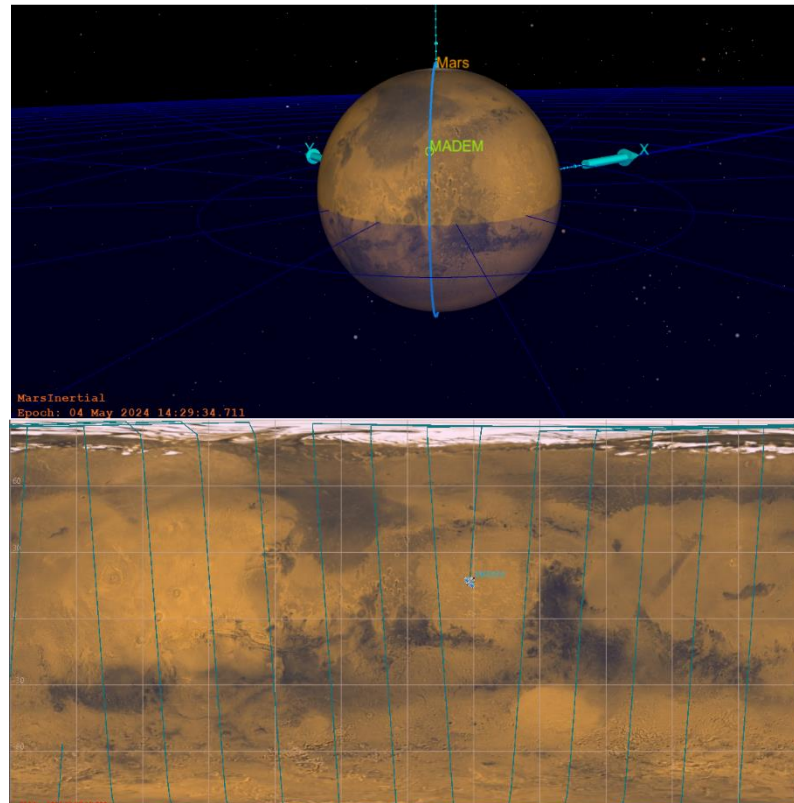


Figure 10: Mars polar orbit, inertial reference frame (above) and ground track (below)

7.3.1 Entry Vehicle Characteristics

In this project, the entry vehicle design is kept out of scope just as the additional properties and equipment used in capsule, including their mass and cost budget.

However, aiming a landing operation on Mars will eventually bring some 6 degree of freedom solutions and flight dynamic issues. Therefore, a conceptual entry vehicle designed is prepared in this manner.

Entry vehicles are the structures that protect the payload and instruments from exceeded heating and enormous dynamic pressures. This is why an approach is developed in a conceptual way. The key factors that are used in EDL techniques can be given as:

- Ballistic coefficient
- Thermal protection system (TPS)
- Ballistic control
- Entry vehicle geometry
- Parachute
- Entry States

In this section, these key factors except entry conditions are given and explained. Entry conditions, however, are given in the next section in detail. Recalling that everything mentioned in this section is out of scope and is given as a conceptual approach. [K6N]

The entry sequence occurs as ballistic entry, where lift to drag ratio is less than one and assumed only weight and drag forces affect the vehicle. In ballistic entry sequence, the trajectory is predetermined by ballistic coefficient, which is given as $m/(C_D S)$ where m is the vehicle mass, C_D is the drag coefficient and S is the reference area. Although drag coefficient is a unitless coefficient and it is a function of the Mach and Reynolds number, assuming a constant value will be enough for predetermining a rough trajectory.

Ablative materials, which change phase during entry and descend sequences, are also used for thermal control purposes. As the material heats up, the solid material starts to vaporize, causing the surface to cool down.

A ballistic control is also needed for moment of inertia, as much as distributing g force. So that the spacecraft will not tumble and get damaged.

As it is aimed to enter the atmosphere via orbit, the use of modified cruise stage is also important; so that the attitude and control of the entry vehicle can be provided. In this conceptual design, the modified cruise stage will provide the required de-orbit burn in specific directions. Figure 7.2 shows the cruise stage of NASA's MER cruise stage, each used in Spirit and Opportunity rovers.

Conceptual geometry of the entry vehicle is given in Figure 7.3, as table 7.1 shows the entry vehicle characteristics.

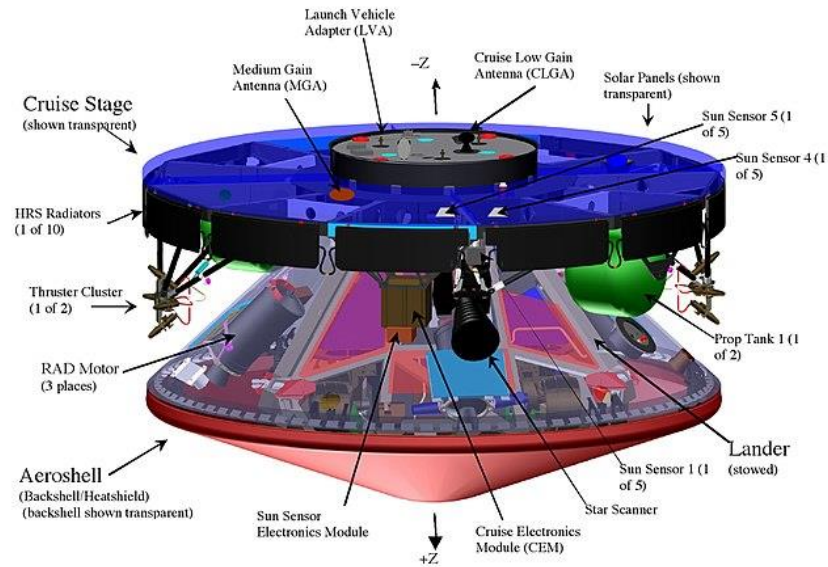


Figure 12: MER Cruise Stage[K7N]

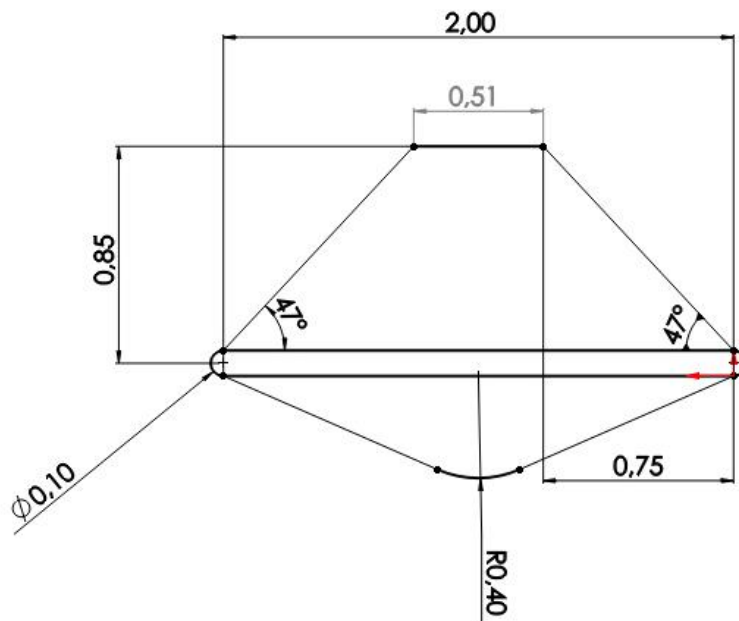


Figure 11:MADEM conceptual capsule design geometry

Table 4: Capsule entry properties

Entry vehicle characteristics	
Ballistic Coefficient	49.74 kg/m ²
Reference Area	3.145 m ²
Control Method	Ballistic, spin stabilization at 2 rpm
Entry Angle	-6° at 3515.2 km radial distance
Parachute	Single disk band gap
Cooling	Ablation
Mass (Spacecraft included)	250 kg

7.3.2 Entry States

To establish an entry state, an appropriate state type must be taken as reference. In this manner, Spherical and Local Horizon based Celestial Coordinate System is used as usual. In this coordinate system, six elements take place just as Cartesian or Keplerian: three for spherical coordinates and three for local horizon. [K8N]

Figure 7.4 shows the frame, and the explanations are given below.

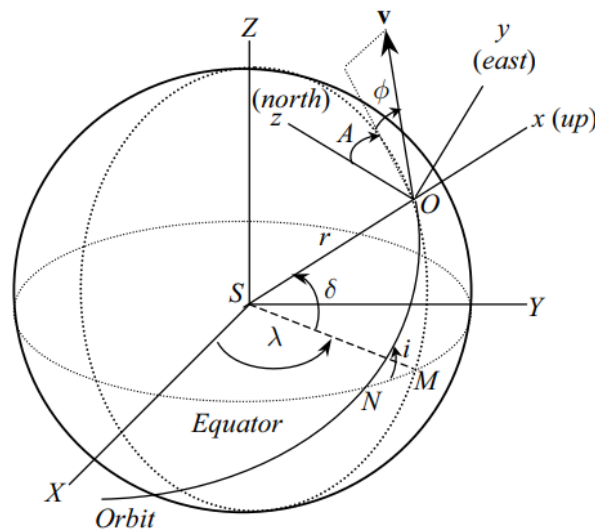


Figure 13: Spherical and Local Horizon based Celestial Coordinate System

Latitude (δ): The angle measured from the equatorial plane, unit degree.

Longitude (λ): The angle measured from X axis to rightward, unit degree.

Radial distance (r): Distance between the planet's center of gravity and the spacecraft itself, unit km.

Velocity (V): Relative magnitude of velocity of the spacecraft, unit km/s

Flight path angle (ϕ): Angle below horizontal of velocity vector in inertial frame. Unit degree and value negative.

Azimuth (A): The angle measured from the north pole in clockwise direction. Unit degree.

Thus, for mission analysis, these elements are considered. For this purpose, a simulation is created in MATLAB. To verify this simulation, a real mission has been tested and verified with related results.

For this issue, MER Spirit's entry states and its conditions until the parachute deployment is chosen and tested. The values are taken from Paul Withers and Michael D. Smith's *Atmospheric entry profiles from the Mars Exploration Rovers Spirit and Opportunity*, published in 2005 and not an open document. Figure 7.5 shows the table for MER Spirit's entry states.

Table 5: Spirit's estimated entry states

Table 2 Estimated entry states with 1- σ uncertainties	
	Spirit
Time— t_{ref} (SCLK ^a s)	2085.625 ± 0
t_{ref} (SCLK s)	126460000.000 ± 0
Radial distance (km)	3522.2 ± 1.7
Areocentric latitude ($^{\circ}\text{N}$)	-17.7 ± 0.04
Areocentric longitude ($^{\circ}\text{E}$)	161.8 ± 0.01
Speed ^b (km s ⁻¹)	$5.63 \pm (7 \times 10^{-4})$
Flight path angle ^c ($^{\circ}$)	11.5 ± 0.02
Azimuth ^d ($^{\circ}$)	79.0 ± 0.02

The states are entered to simulation as input, whose block diagram is shown in figure 7.6

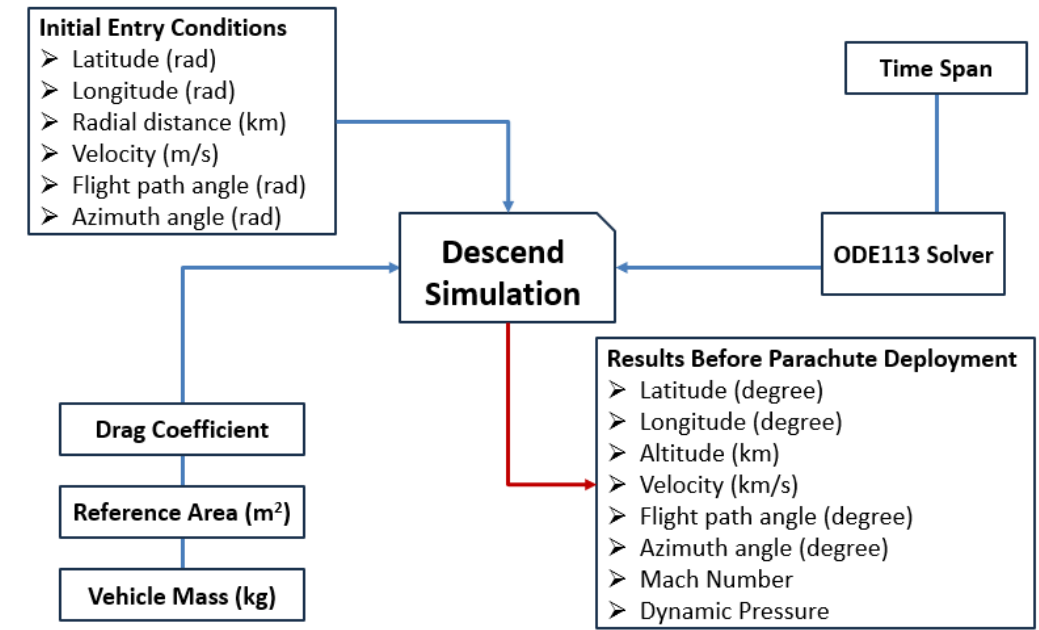


Figure 14: Descend Simulation Block Diagram

Note that the simulation is valid only up to parachute deployment. Figure 7.7 shows us the simulation results done in MATLAB, while Figure 7.8 shows conditions at parachute deployment, with uncertainties.

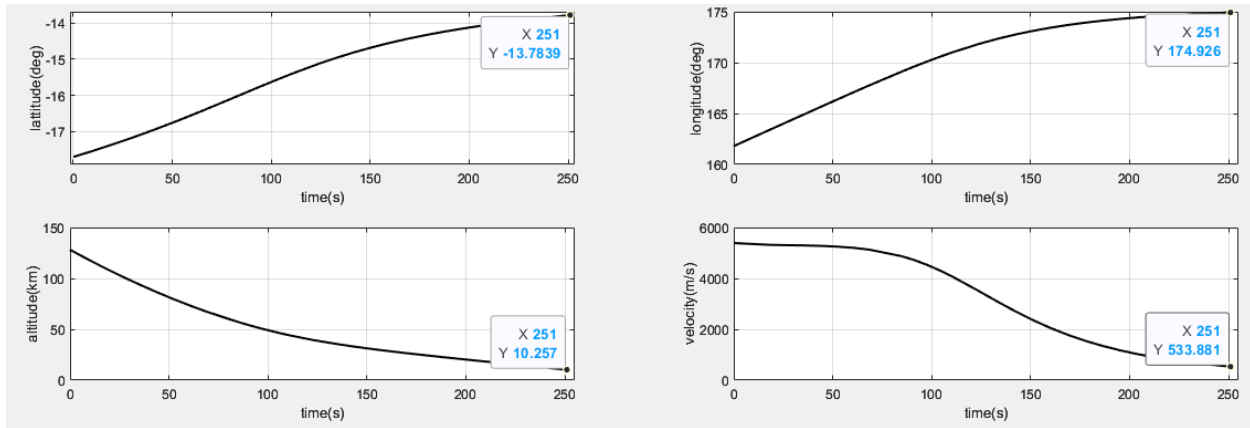


Figure 15: Descend Simulation

Table 6: Spirit's conditions at parachute deployment

Table 3 Conditions at parachute deployment with 1- σ uncertainties	
	Spirit
Time— t_{ref} (SCLK s)	2336.375 ± 0
Altitude (km)	7.5 ± 1.7
v_{rel} (m s^{-1})	410.98 ± 0.77
Latitude ($^{\circ}\text{N}$)	-14.528 ± 0.039
Longitude ($^{\circ}\text{E}$)	175.411 ± 0.013

Table 7: Simulation Results Comparison

Parameter	Simulation Result	Desired Value	% Error
Latitude (degree)	-13.7839	-14.489	4.866
Longitude (degree)	174.926	175.398	0.269
Altitude (km)	10.257	9.2	11.489
Velocity (m/s)	533.881	411.75	29.661

Note that higher errors are encountered in velocity and altitude. But in ballistic atmospheric entry dynamics, altitude and velocity are not possessing a strong importance after the maximum deceleration is achieved, since the vehicle will get to its terminal velocity after parachute is deployed no matter what its beforehand velocity was. Plus, the maximum deceleration causes spacecraft or any other body, such as meteor, to descend relatively more vertical line, meaning not much difference will occur in longitude and latitude [K9N] as it is shown in figure 7.10, which is from *NASA's Mars Exploration Rovers EDL Trajectory and Atmosphere Reconstruction Using NewSTEP*, which draws a similar path to figure 7.7.

Therefore, the simulation presented can be used for this project.

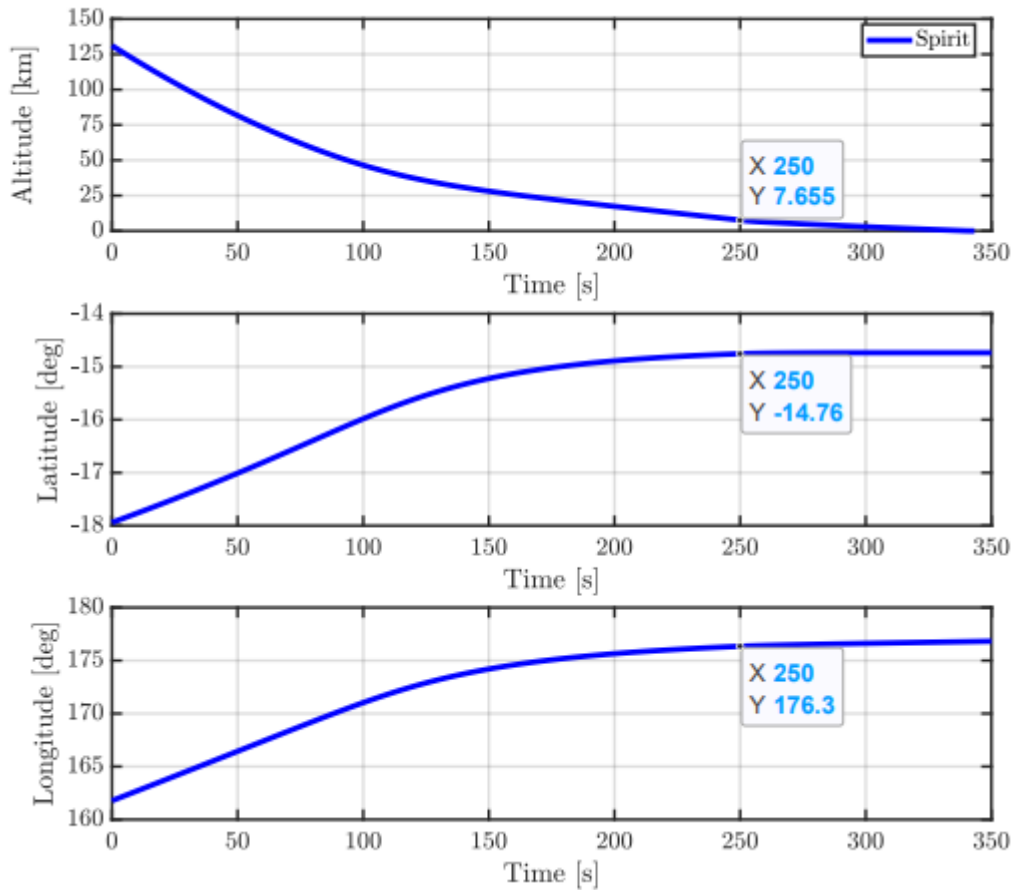


Figure 16: Reconstructed trajectory of MER Spirit

7.3.3 Descend

Table 8: MADEM Entry State

Latitude	0.5°
Longitude	174.2°
Radial Distance	3515.2 km
Relative Velocity	2953.665 m/s
Flight Path Angle	-6°
Azimuth	180°

With numerous interpolations, the entry state of the MADEM is determined as it is shown in table

The same simulation is used until MADEM's parachute deployment occurs, which takes 250 seconds from the beginning of the entry. Figure 7.11 shows the descend panel of MADEM. Note that the altitude is based on Mars' Sea level rather than landing point, meaning it has less distance to touch down than it is shown.

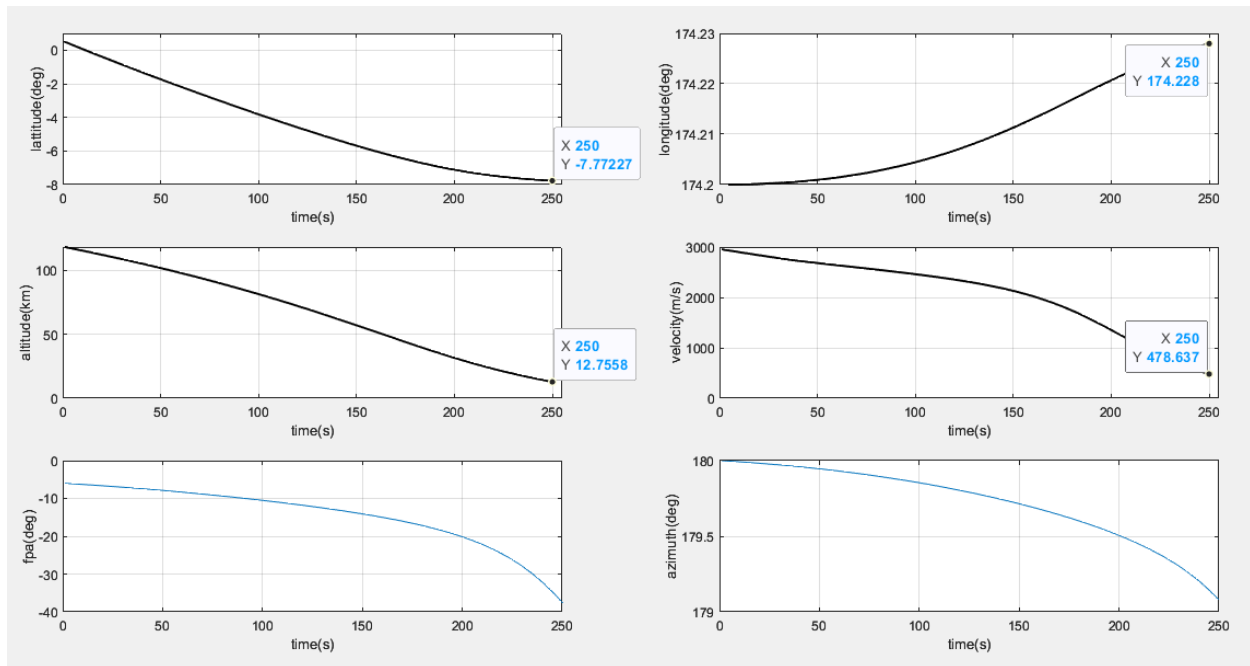


Figure 18:MADEM Descend trajectory

In the same manner, the MATLAB codes are integrated into Google Earth. Note that the vehicle descends from the Northern side and the time is set to 12.00 am UTC, which is daytime on Mars and required for the use of visual based navigation.

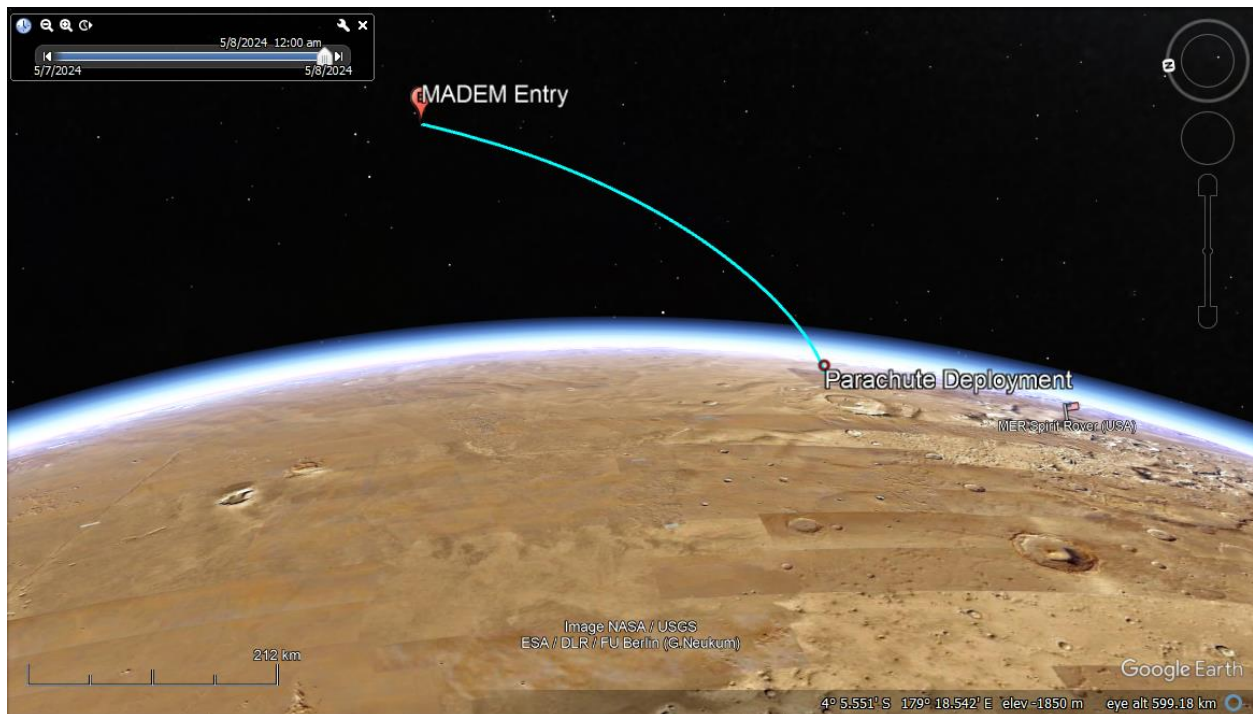


Figure 17: MADEM descend panel

7.3.4 Parachute Deployment

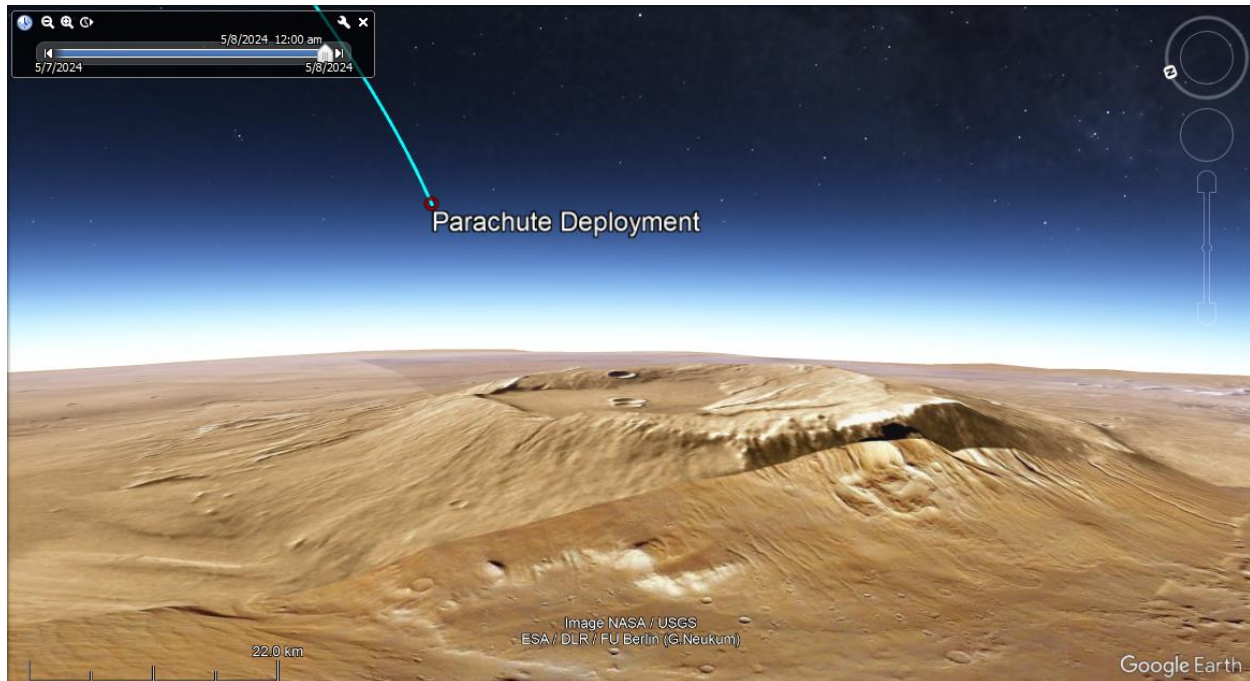


Figure 19:MADEM Parachute Deployment

According to the simulation, there is roughly 10.5 km distance between the parachute deployment altitude and the landing point elevation, considering northern elevation of the mountain to be around 2.2 km.



Velocity During Recovery

Glenn
Research
Center

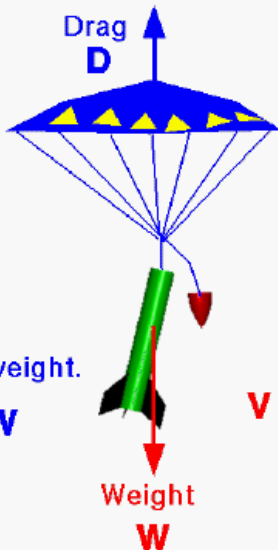
C_d = drag coefficient of parachute
 $C_d = 1.75$ (typical)

Drag Equation:

$$D = C_d \frac{\rho V^2}{2} A$$

During recovery, drag=weight.

$$D = C_d \frac{\rho V^2}{2} A = W$$



ρ = air density
 $\rho = 1.229 \text{ kg / cu m}$
 A = parachute area
 V = velocity

Solve for Velocity:

$$V = \sqrt{\frac{2 W}{C_d \rho A}}$$

Figure 20:Velocity after parachute[K10N]

Figure 7.11 shows how to find the velocity after parachute deployment. In this manner, some appropriate assumptions are considered. So that;

- The DGB parachute's drag coefficient is 0.6 and diameter is 10 meters.
- The vehicle's 250 kg mass is constant, although ablation and heat shield separation occur.
- Deployment occurs at 12.76 km altitude from the sea level, where the density is 0.00521 kg/m^3

Using the equation given in Figure 7.11, the velocity of MADEM after parachute deployment is found to be 86.63 m/s. To compare, EDM's velocity before and after is taken from its descent sequence [K10N]

Velocity	EDM Schiaparelli	MADEM
Before the deployment	472.22 m/s	478.63 m/s
After the deployment	88.89 m/s	86.63 m/s

However, one should note that the density is increasing as the altitude is decreasing. Therefore, the velocity when the spacecraft separates the capsule won't be the same as the velocity at parachute deployment.

As mentioned previously, the caldera of the Apollinaris is about 2.2 km high from the sea level. Separating spacecraft 2 km above the caldera can be considered to be enough to land successfully with the help of ignitors. To do that, the parachute should be coasting 8.3 km vertically.

The density at the 4.2 km altitude, where the thrust ignition starts, is about 0.0107 kg/m^3 , which slows the entry vehicle down from 86.63 m/s to 60.7 m/s. Thus, it can be assumed that the average speed between the parachute deployment and the spacecraft separation is 73.65 m/s.

This average speed gives the time passed, as we have vertical distance of 8300 m and velocity of 73.65 m/s, resulting in 113 seconds, which is quite enough for heat shield separation, radar and visual based navigation activation and spacecraft separation.

7.3.5 Landing

For landing phase, 1 dimensional dynamic equation are used in MATLAB. Igniting thrusters for 33 seconds, providing 800N force upwards as total will lead spacecraft land safely.

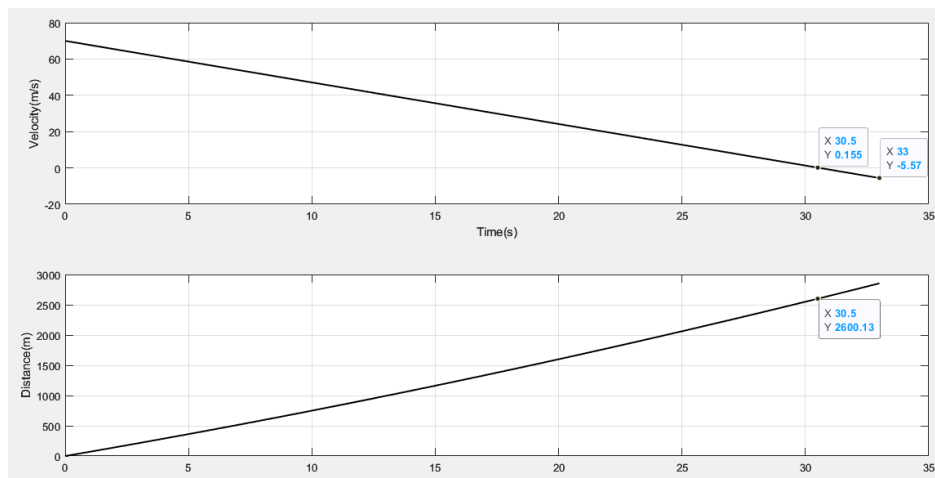


Figure 21:MADEM touchdown

Figure 7.12 shows the touchdown panel. The spacecraft reaches 0 m/s at around 31st, meaning it is ready to turn off and touchdown. The exceeded 5.57 m/s is the delta V margin that can be used for guidance control. The negative sign states that it starts going upward, which is not going to be used for it.

Figure 7.13 shows the propulsion diagram.

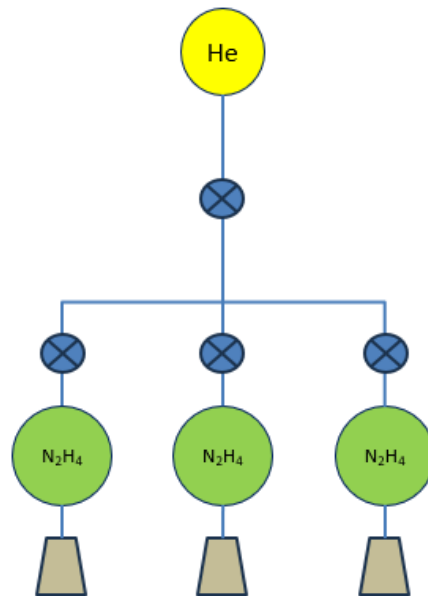


Figure 22:MADEM propulsion diagram

For the propulsion system, MADEM is considered to have.

- 3x CHT-400N monopropellant engines, each producing up to 440N thrust.
- 3x 6.29 Liters of propellant tanks each having 5 kg of hydrazine.
- One pressurant tank having Helium.
- Pressure regulators and various valves (such as pyro, fill-drain, and thruster valve) for propulsion control.

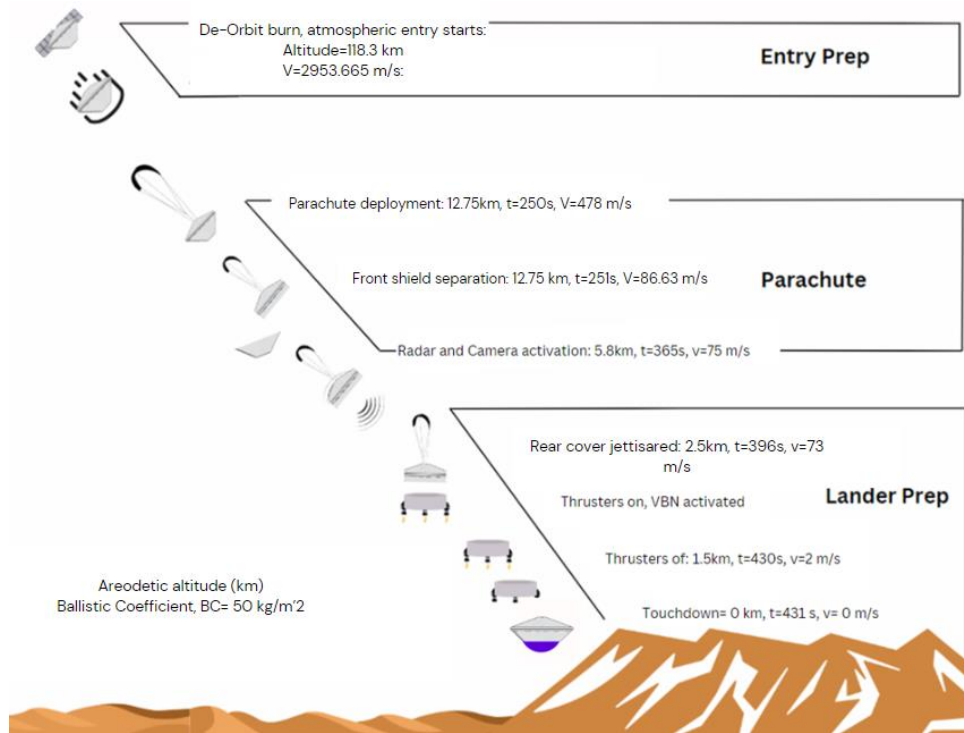


Figure 24:CONOPS

Table 9:Delta-V

	ΔV (km/s)
De-orbit burn	3.4786
Aerocapture	306.074
Thruster ignution	0.07
Total	34.786

SINIR

7.4 Platform Trade-offs

On-board computer trade-offs

While choosing the landers on-board computer, which is the brain of the lander, we turned our attention to the four main requirements.

These are;

- Technology readiness level (TRL)
- Required RAM
- Required memory
- Common operating system
- Past mars missions

Within these requirements, we made a trade-off between most reliable onboard computers, as follows:

Table 10:Communication Subsystem Trade-offs

CPUs	RAM (MB)	Memory (MB)	Operating System	Mission
2 MHZ 8-bit Intel 80C85	0.5	0.172	Cylic Executive	Sojourner rover
20 MHZ 32-bit BAE RAD 6000	128	256	VxWorks	Curiosity Rover
200 MHZ 32-bit BAE RAD 750	256	2048	VxWorks	Perseverance

7.4.1.1 Communication Subsystem trade-offs

7.4.1.1.1 Rf band trade-off

Since the lander will land on Mars, all bands were available to be used for communication with relay satellites. But based on the Signal-to-noise ratio (Eb/No), Bandwidth, Attenuation, Data rate, the S band chosen for the lander inter communication.

Table 11:Band Trade-Offs

Band	Frequency (GHz)	Comment
X-band	8.0-12.0	Provides balance between data rate, antenna size, and atmospheric attenuation.

S-band	2.0-4.0	TRL is very high. Provides reliable communication. Less susceptible to atmospheric absorption. Mostly used for military communications.
Ka-band	26.5-40.0	Provides higher data rate. Larger antenna required for atmospheric attenuation.
UHF	0.3-3	Provides less data rate. Better penetration through atmosphere. Suitable for tracking, command, and telemetry.

Antenna Trade-off

The antenna has been made from data rate, needed gain, total mass, power consumption, and past usage (TRL). From trade-off analysis, parabolic dish has been chosen to be the most efficient, reliable fit for the mission.

Table 12:Antenna Trade-offs

Antenna	Flight History	Comment
Optical	Limited	Achievable high data rates. Very precise pointing is required.
Phased Array	Some	High data rates possible. No need for a mechanical structure for the pointing. Keeps small area. Very fast and precise steering of the communications beam.
Omni-directional	Extensive	360 degrees of coverage. No pointing is required. Low data rates are possible. High power is needed to achieve high data rates.

Parabolic dish	Extensive	Exceptionally reliable. Provides high profits. Consumes less power. Must be mechanically pointed. Keeps large area.
----------------	-----------	--

Power system trade-off

The MADEM lander mission aimed to have almost half mars day (sol) solar radiation to make the mission feasible. From this reason, solar-powered panels considered to be the best option for power system.

Table 13:Power subsystem Trade-offs

Power System Type	Advantages	Disadvantages
Solar-powered panel	Renewable energy source. Lower cost than RTGs. Size and power changes with mission requirements. Reliable technology (TRL is high).	Depends on the sunlight availability. Martian dust can reduce panel efficiency. Extra dust-tolerant design needed. High powered missions require large, deployable panels. Not suitable for low-light environment.
Radioisotope Thermoelectric Generator (RTG)	No need for sunlight. Long lifespan provided by radioactive materials. Compact and lightweight. No external arts (like cleaning mechanisms) are required.	High cost. Provides lower power compared to large solar arrays. Disposal is a key factor. Verification of RTGs can be difficult.

Figure –5: CONOPS

Table 14:Delta-V

The CONOPS diagram you provided illustrates the descent of a parachute. The diagram shows the parachute's journey through the atmosphere using altitude in kilometers and time in seconds. Each significant event is marked with a text label and an icon on the diagram. The main events are as follows:

- **Atmospheric Entry:** The parachute enters the atmosphere at an altitude of 125.2 kilometers and a speed of 3,478.6 meters per second.
- **Preparation for Entry:** The parachute prepares for deceleration in the atmosphere.
- **Start of Atmospheric Deceleration:** The parachute begins to decelerate at an altitude of 81.5 kilometers and 102 seconds later, at a speed of 3,130.7 meters per second.
- **Parachute Deployment:** The parachute deploys at an altitude of 6.4 kilometers and 375 seconds later, at a speed of 305 meters per second.
- **Separation of the Front Shield:** The parachute's front shield separates at an altitude of 5.8 kilometers and 365 seconds later, at a speed of 75 meters per second.
- **Radar and Camera Activation:** The parachute's radar and camera are activated at an altitude of 5.8 kilometers and 365 seconds later, at a speed of 75 meters per second.
- **Rear Cover Ejection:** The parachute's rear cover is ejected at an altitude of 3.5 kilometers and 396 seconds later, at a speed of 73 meters per second.
- **Thrusters On, VBN Active:** The parachute's thrusters turn on at an altitude of 1.5 kilometers and 430 seconds later, at a speed of 2 meters per second, and switch to vertical landing mode.
- **Landing:** The parachute lands on the ground at an altitude of 0 kilometers and 431 seconds later, at a speed of 0 meters per second.

On the right side of the diagram, the parachute's ballistic coefficient (40 kg/m^2) is shown. The ballistic coefficient is a measure of an object's resistance to air. The higher an object's ballistic coefficient, the higher its air resistance.

This CONOPS provides a simplified overview of a parachute descent. A real parachute descent can be much more complex than what is shown in this diagram. For example, wind direction and speed can affect the parachute's descent path.

7.5 Technical Solution

The product tree for our Mars Lander is meticulously designed to ensure all systems and subsystems are clearly organized and functionally integrated. The primary components and their respective subcomponents are outlined as follows:

7.5.1.1 *Lander*

The mission's core, the Mars Lander, is designed to perform a variety of scientific and exploratory tasks once it reaches the Martian surface. The Lander is subdivided into several key systems, each responsible for different functionalities.

7.5.1.2 *Communication*

- **Transmitter**
- **Receiver**

The communication system is crucial for maintaining contact with mission control on Earth. It consists of a transmitter and a receiver, enabling the Lander to send and receive data.

7.5.1.3 *Power*

- **Li-Ion Battery**
- **Power Control and Distribution Board**
- **Solar Panel**

The power system ensures that the Lander has a reliable energy source. It includes lithium-ion batteries, a power control and distribution board, and solar panels to harness solar energy.

7.5.1.4 *Command and Data Handling*

- **Single-Board System**

This subsystem handles the Lander's computational and data processing needs. It comprises a single-board system that serves as the brain of the Lander, processing data and executing commands.

7.5.1.5 *Attitude Determination and Control System*

- **Accelerometer**
- **Gyroscope**
- **Sun Sensor**
- **Torque Sensor**

This system is responsible for maintaining the Lander's orientation and stability. It includes accelerometers, gyroscopes, sun sensors, and torque sensors to monitor and adjust the Lander's attitude.

7.5.1.6 *Science Payload*

- **Panoramic Camera**
- **Nadir Pointing Panchromatic Camera**
- **Robotic Arm with LIBS**
- **Magnetometer**

- **Seismometer**

The science payload consists of various instruments for conducting scientific research. These include a panoramic camera, a nadir pointing panchromatic camera equipped with a robotic arm and Laser-Induced Breakdown Spectroscopy (LIBS), a magnetometer, and a seismometer for geological studies.

7.5.1.7 Thermal

- **Temperature Sensor**
- **Thermostat**

The thermal system ensures the Lander's internal temperature is maintained within operational limits. It includes temperature sensors and thermostats to regulate the Lander's thermal environment.

Each subsystem is carefully integrated to ensure the Lander operates efficiently and effectively on the Martian surface. This product tree provides a comprehensive overview of the Lander's design and the intricate interplay of its components, all of which are crucial for the success of the mission.

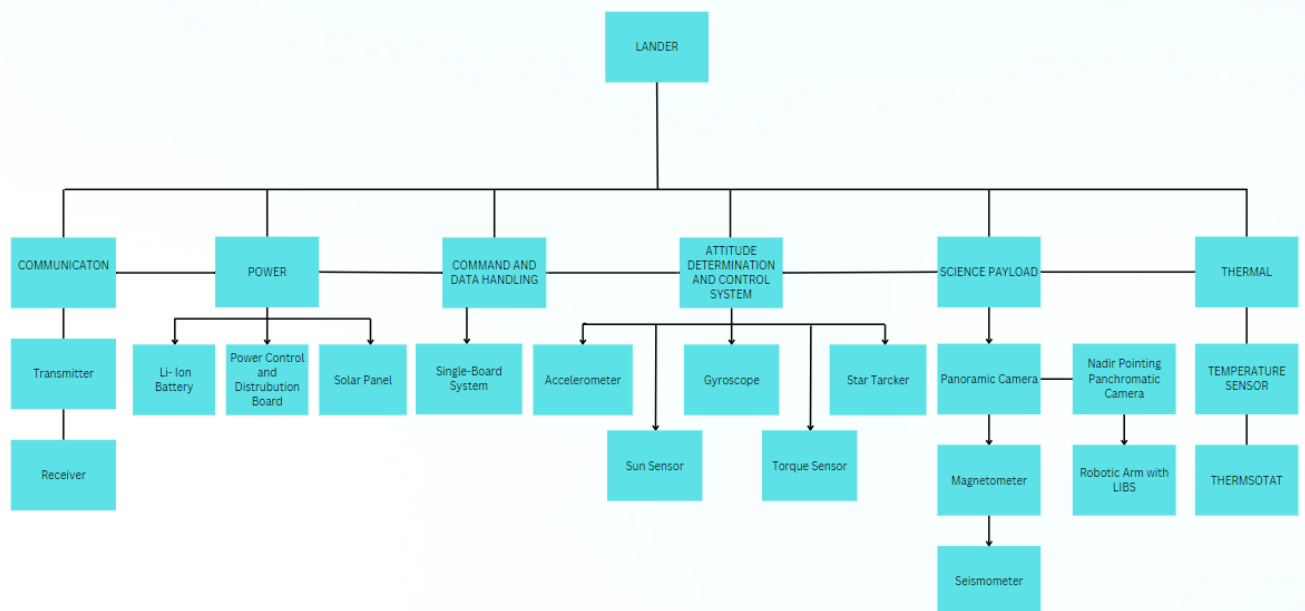


Figure 25:Product Tree

The Payload

Computer (RAD 750):

- Model/Specifications: RAD 750, featuring a 200 MHz 32-bit processor.
- Supplier: BAC SYSTEMS.
- Description: This single-board, panoramic computer serves as the Mars Rover's brain. It processes all scientific data and controls commands. RAD 750 is designed to meet reliability and durability requirements in space environments.

Panoramic Camera (SLDU 345):

- Model/Specifications: SLDU 345 captures panoramic images of Mars at 2,400xGSD m resolution (1/4 for multispectral).
- Supplier: CAVATOSRASS UK Ltd.
- Description: This camera captures panoramic images of Mars to learn about its geology and terrain. It allows for multispectral capability and provides high resolution.

PANCAM (Mastcam-Z):

- Model/Specifications: Mastcam-Z is a dual camera system with a resolution of 600 x 1200 pixels and a field of view of 34" to 63° wide angle and 6° to 15° narrow angle.
- Supplier: Malin Space Science Systems (MSSSI).
- Description: This dual camera system is used to examine close-ups and textures of rocks and soil on Mars. It provides detailed images at different angles.

Robotic Arm (Low-power arm for sample collection):

- Model/Specifications: This arm is used to collect samples of rocks and soil from Mars.
- Supplier: To be designed in-house.
- Description: The robotic arm is designed to collect samples from Mars for further analysis on Earth.

LIBS Instrument (ChemCam):

- Model/Specifications: ChemCam analyzes the chemical composition of rocks and soil using laser induced breakdown spectroscopy (LIBS).
- Supplier: LANL and CNES.
- Description: ChemCam provides information about Mars' geological history by analyzing the chemical composition of rocks and soil.

Magnetometer (Fluxgate Mag-13):

- Model/Specifications: Fluxgate Mag-13 is used to learn about Mars' gravity field and magnetosphere.
- Supplier: Bertington Instruments.

- Description: This magnetometer measures Mars' magnetic field and gravity to understand its magnetosphere.

Seismometer (FRIPPON):

- Model/Specifications: FRIPPON (French Seismometer for Internal Structure) investigates Mars' internal structure and seismic activity.

- Supplier: CNES.

- Description: This seismometer is used to study Mars' internal structure and underground activity.

Transmitter (5-Band TTC Transmitter STC-M503):

- Model/Specifications: 5-Band TTC Transmitter communicates command, control, and telemetry (TTC) data back to Earth.

- Supplier: Honeywell.

- Description: This transmitter sends data collected by the Mars Rover back to Earth for analysis.

VPCDU-1 (General model PODU):

Model/Specifications: General model PODU controls all Mars Rover's electrical systems.

- Supplier: Vectranic.

- Description: This power distribution and control unit (PDCU) manages and controls the electrical systems of the Mars Rover.

Battery (Lithium-ion Range Safety Battery PN 39401 28.8 V):

- Model/Specifications: This lithium-ion battery powers the Mars Rover.

- Supplier: Space Vector.

- Description: This battery provides power to the Mars Rover.

Solar Cell (Nominal (33.6 V to 20.0 V) Gallium Arsenide Solar Cell 30.15 cm):

- Model/Specifications: These 56498-gallium arsenide solar cells provide solar energy to power the Mars Rover.

- Supplier: Unspecified.

- Description: These solar cells collect solar energy to power the Mars Rover.

Thermostats (High Reliability Hermetic Thermostats):

- Model/Specifications: These thermostats control the Mars Rover's thermal system to prevent it from overheating.

- Supplier: Unspecified.

- Description: These thermostats regulate the temperature of the Mars Rover's components.

Temperature Sensors (Model 0118MF):

- Model/Specifications: These temperature sensors monitor the temperature of the Mars Rover's components to ensure they operate within safe limits.
- Supplier: Collins Aerospace.
- Description: These temperature sensors ensure that the Mars Rover's components operate within safe temperature ranges.

Gyroscope (BND085):

- Model/Specifications: This gyroscope tracks the Mars Rover's orientation and rotation to help it navigate.
- Supplier: BOSCH.
- Description: This gyroscope is used to track the orientation and rotation of the Mars Rover for navigation purposes.

Mechanical Design

[Describe your general arrangement.

Go over the check list from the textbook. Be sure the FoVs are not blocked.

Choose structure types.

Show your CAD model.

Describe various configurations (launch configuration, etc.)

Describe mechanisms.]

This is the general design of MADEM. There is robotic arm which is going to leave the seismometer on the ground and make sample collection.

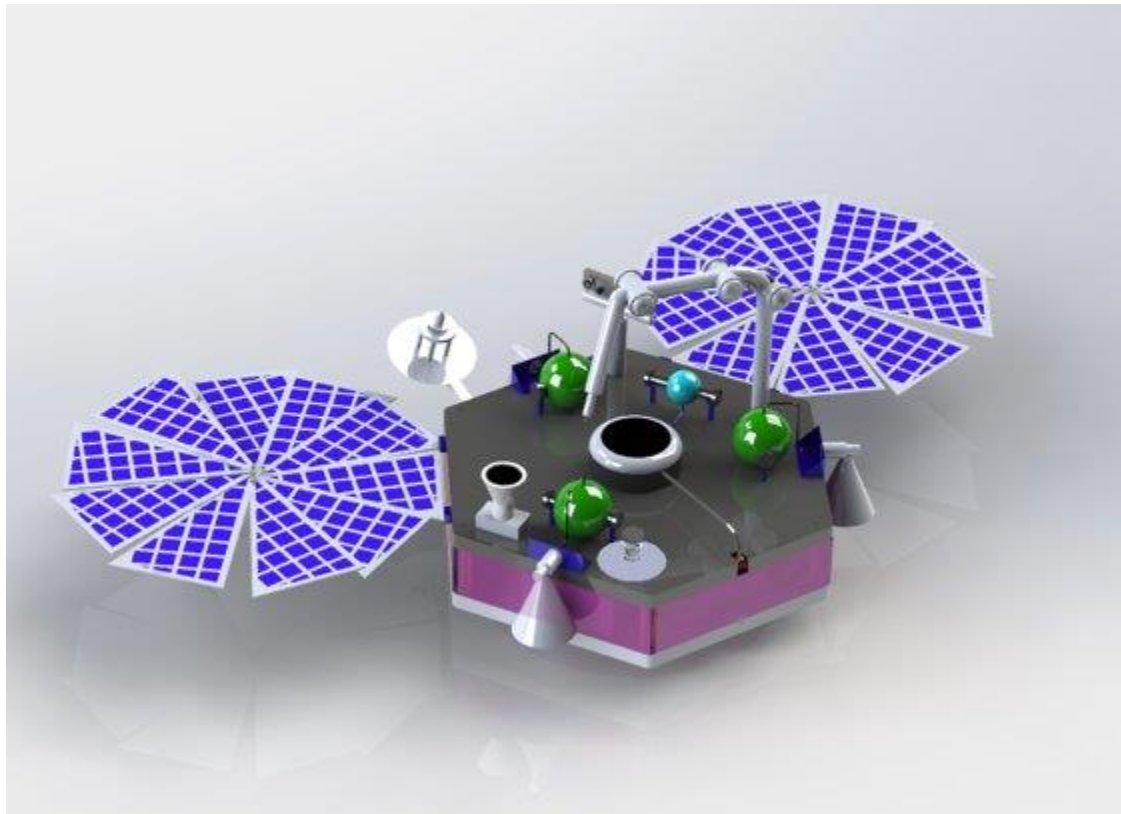


Figure 26: CAD view of the spacecraft

Figure .14. CAD view of the spacecraft. (Example from <https://pressbooks-dev.oer.hawaii.edu/epet302/chapter/4-4-general-arrangement-and-design-drivers/>)

This spacecraft has 2 hexagonal shaped main decks. Since it is strong and lightweight, Aluminum hardened epoxy-carbon fiber used for touchdown phase on the bottom deck.

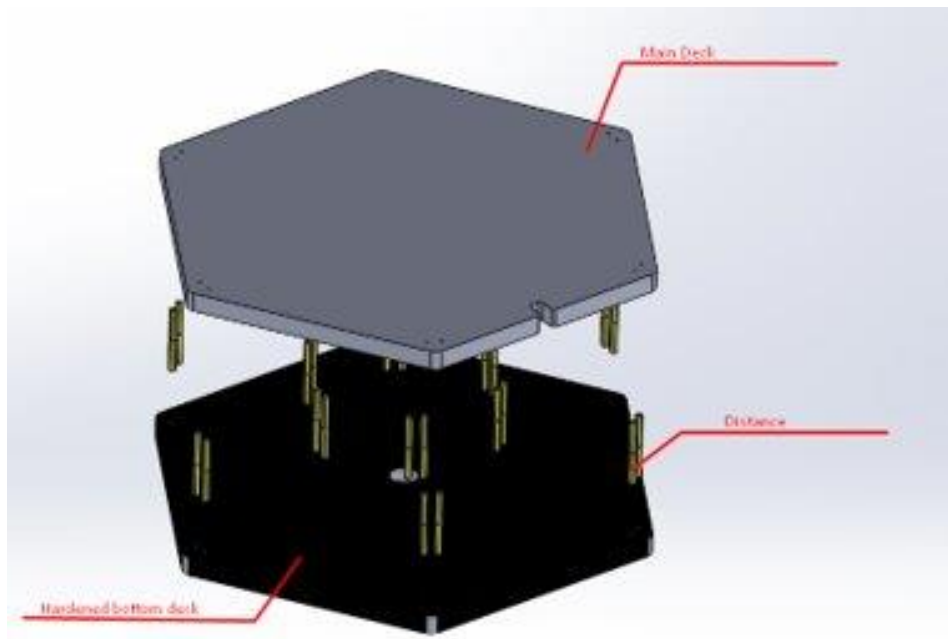


Figure 27:CAD view of the spacecraft-2

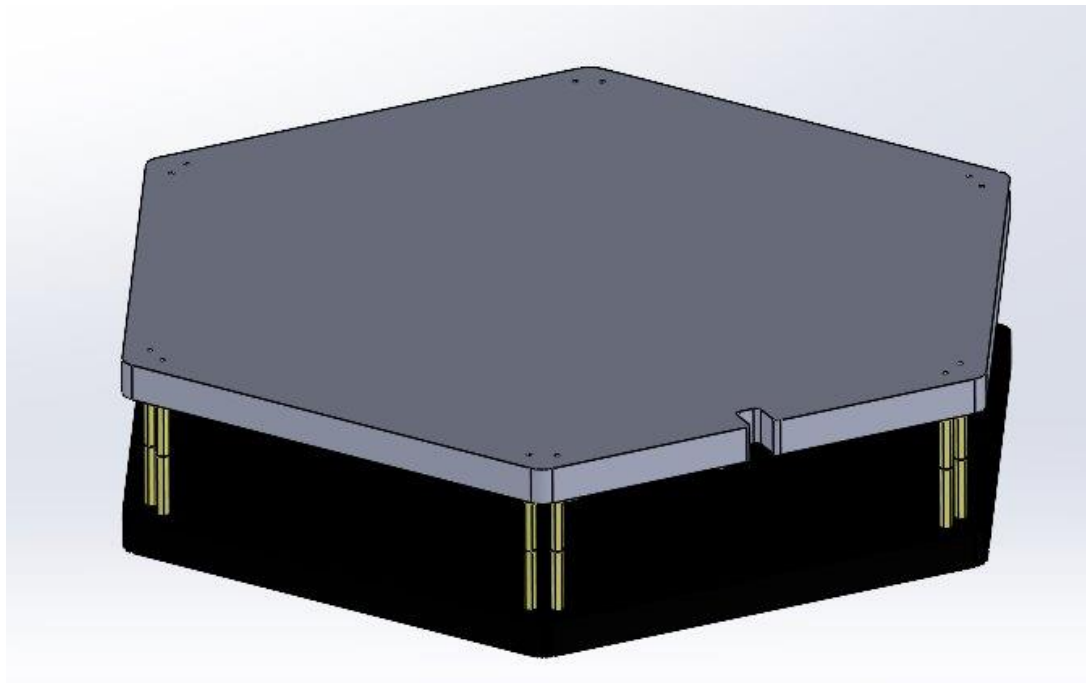


Figure 28:CAD view of the spacecraft-3

For a reliable landing, this spacecraft uses a honeycomb. Inside the honeycomb there is a landing camera. With this camera we can observe the place where we are going to land.

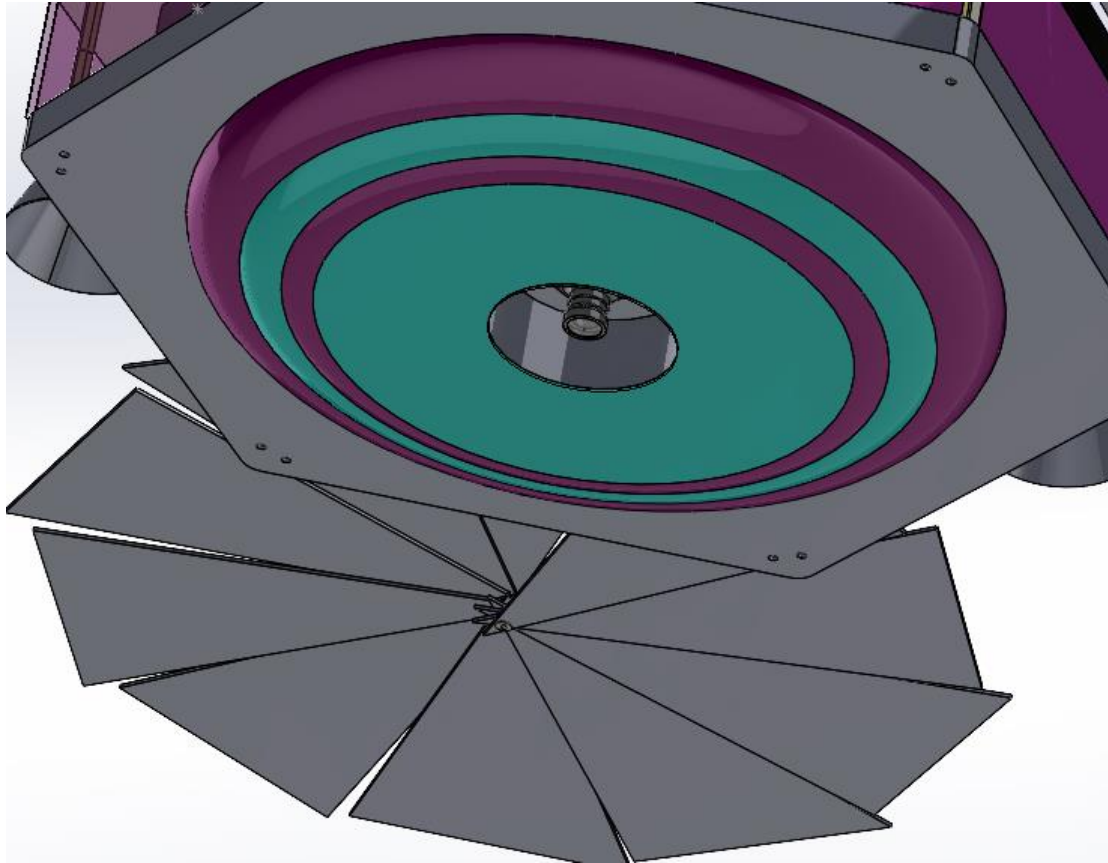
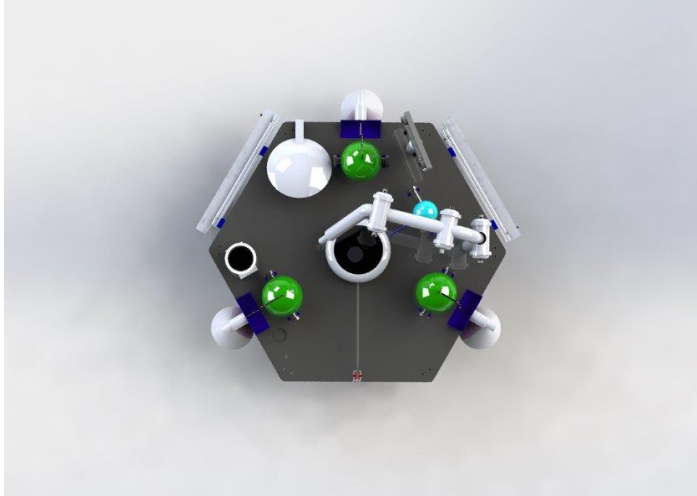


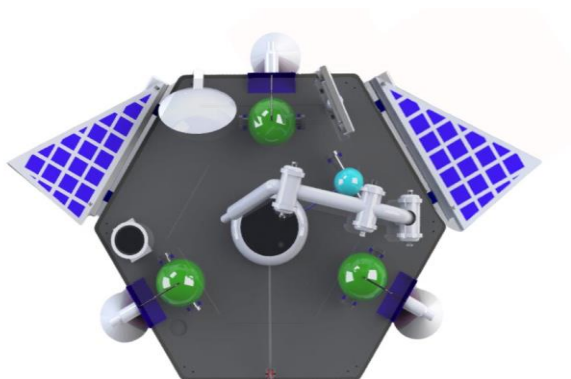
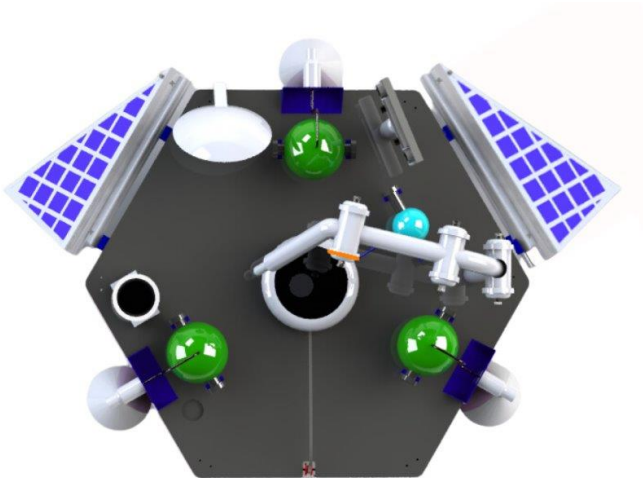
Figure 29:CAD view of the spacecraft-3

MADDEM has 2 equal and symmetrical solar panels which are foldable and inspired by the design of a fan. When the MADDEM lands solar panels will be opened gradually. Each solar panel has 1,818 meters square area. Total area of the solar panels 3,636-meter square. You can see the opening mechanism of the solar panels in the figures.

I)



II)



III)

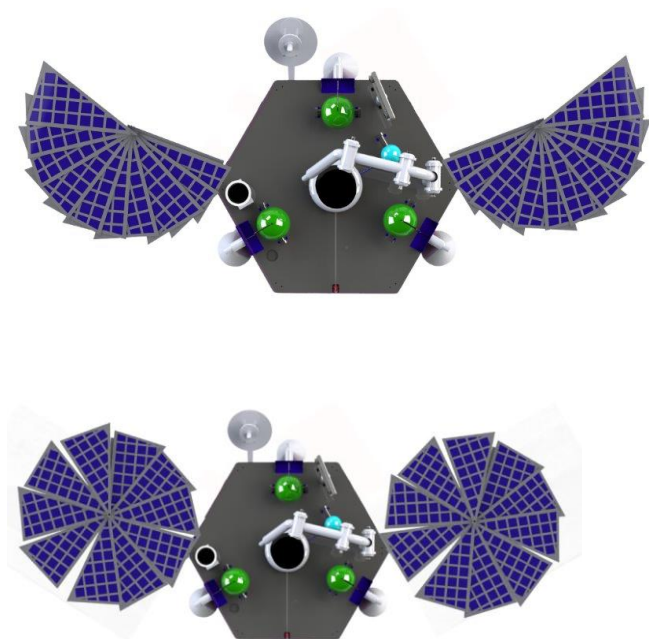


Figure 30:CAD view of unfolding solar panel

Electronic Design

[Decide on the equipment you will use. Give the name, model, supplier etc. for each equipment. Show them as a table. Ad a hyperlink to the supplier page.]

Table 15:The equipment used in the spacecraft

The Equipment	Model/properties	Supplier	# of units
ON-BOARD COMPUTER	200 MHz 32-bit RAD 750 Single-board	BAE SYSTEMS	2
Panchromatic camera	S.LDU 345	CAVU Aerospace UK Ltd	1

	2,400xGSD m (1/4 for multispectral)		
PANCAM	Mastcam-Z 600 x 1200 pixels 34° to 63° for wide-angle and 6° to 15° for narrow angle	Malin Space Science Systems (MSSS)	2
Robotic arm	Low-Power arm (for sample collection)	To be designed in-house	1
LIBS instrument	ChemCam	LANL and CNES	1
Magnetometer	Fluxgate Mag-13	Bartington Instruments	2
Seismometer	FRIPPON (French seismometer for Interior Structure)	CNES	1
Transceiver	S-Band TTC Transceiver STC-MS03	Honeywell	1
VPCDU-1	Generic model PCDU	Vectronic	1
Battery	Lithium-ion Range Safety Battery PN 39401 28.8 V Nominal (33.6 V to 20.0 V)	Space Vector	4
Solar Cell	Gallium Arsenide Solar Cell 30.15cm ²	CETC	56498
Thermostats	High Reliability Hermetic Thermostats Switch type = SPST	Honeywell	1
Temperature Sensors	Model 0118MF	Collins AeroSpace	1
Gyroscope	BNO085	BOSCH	1
Sun Sensor	50 mW 12 to 50V	SITAEL	1
Torque Sensor	T25	Interface Inc.	1
Accelerometer	Three Axis Accelerometer	Airbus	1
Star Tracker	ST200	AAC Clyde Space	1

Mars Lander Block Diagram Overview

The block diagram provides a detailed view of the interfaces and interconnections between various subsystems of the Mars Lander. The different lines indicate specific types of connections: power, data, RS422, I2C, SPI, and UART. Below is a detailed explanation of the subsystems and their interconnections:

1. Communication

- Transmitter

-Receiver

The communication system is linked to the Command and Data Handling system using RS422 (red lines) for data transmission. This ensures reliable and robust data exchange with mission control.

2. Power Systems

- Solar Array
- Lithium-ion Battery
- Power Control and Distribution Board

The power systems generate and distribute electrical power. The Solar Array and Lithium-ion Battery feed into the Power Control and Distribution Board, which then distributes power to all other subsystems (represented by red lines).

3. Command and Data Handling

- Single-board System

The single-board system acts as the central processing unit, interfacing with all other subsystems through various protocols. It communicates with:

- Communication (RS422, red lines)
- Science Payloads (RS422, I2C, SPI, and UART - a combination of black, blue, purple, and green lines)
- Attitude Determination and Control System(RS422, I2C, and SPI)
- Thermal (I2C and SPI)

4. Attitude Determination and Control System

- Gyroscope
- Accelerometer
- Sun Sensor
- Torque Sensor
- Star Tracker

These components are crucial for maintaining and controlling the lander's orientation. They are connected to the Single-board System using RS422, I2C, and SPI protocols, ensuring accurate and real-time data transmission.

5. Science Payloads

- Panoramic Camera
- Nadir Pointing Panchromatic Camera
- Robotic Arm with LIBS

- Magnetometer
- Seismometer

These instruments collect scientific data from the Martian surface. Each component interfaces with the Single-board System:

- Panoramic Camera and Magnetometer via RS422 (red lines)
- Nadir Pointing Panchromatic Camera and Robotic Arm with LIBS via I2C (blue lines) and SPI (purple lines)
- Seismometer via RS422 and I2C

6. Thermal

- Temperature Sensor
- Thermostat

The thermal subsystem maintains the operational temperature of the lander. The Temperature Sensor and Thermostat are connected to the Single-board System using I2C and SPI protocols (blue and purple lines).

Interconnections:

- Power Lines (Red): These supply power from the Power Control and Distribution Board to all subsystems.
- Data Lines (Black): These are used for general data communication.
- RS422 (Red): A robust communication protocol for long-distance data transfer, used for critical connections such as between the communication system and the command and data handling system.
- I2C (Blue): A protocol for communication between integrated circuits, used for connecting various sensors and instruments.
- SPI (Purple): A high-speed communication protocol used for short-distance data transfer.
- UART (Green): A protocol for asynchronous serial communication, used for simpler and slower data transfer needs.

This block diagram illustrates the intricate network of connections that enable the Mars Lander to function as a cohesive unit. Each subsystem is carefully integrated, ensuring reliable power distribution, data handling, and communication, all critical for the successful operation of the lander on Mars.

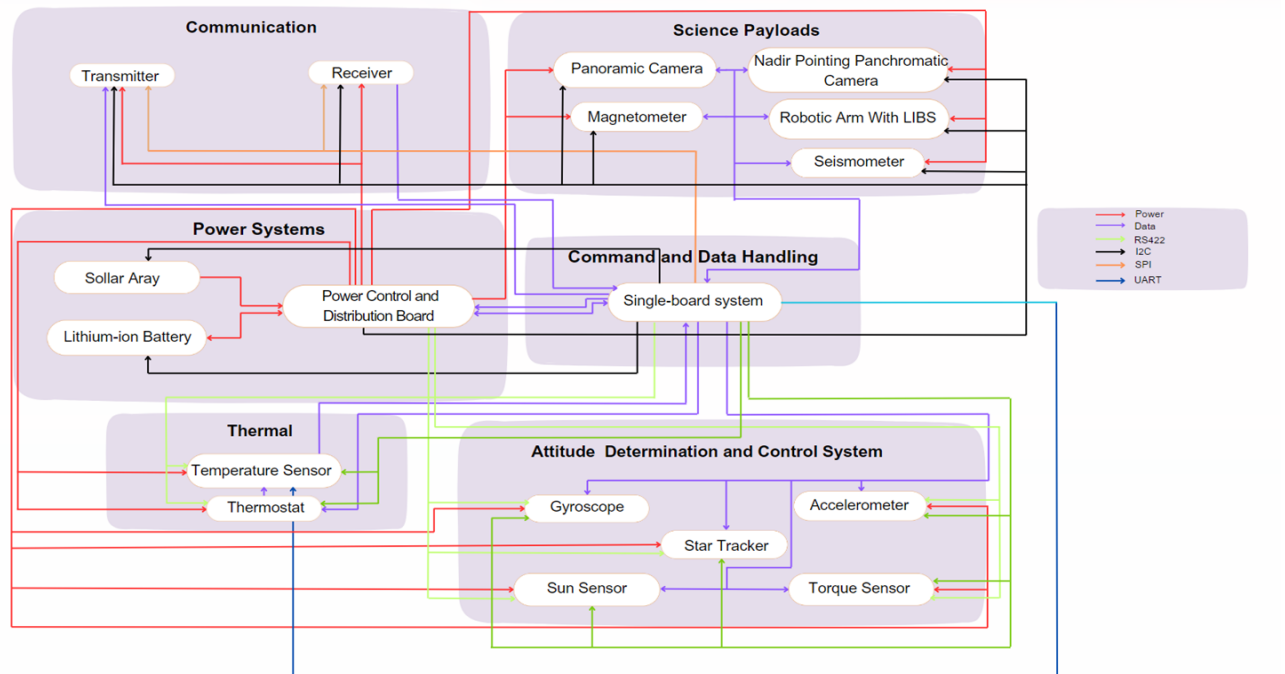


Figure 31: The block diagram of the system

Figure .15. The block diagram of the system.

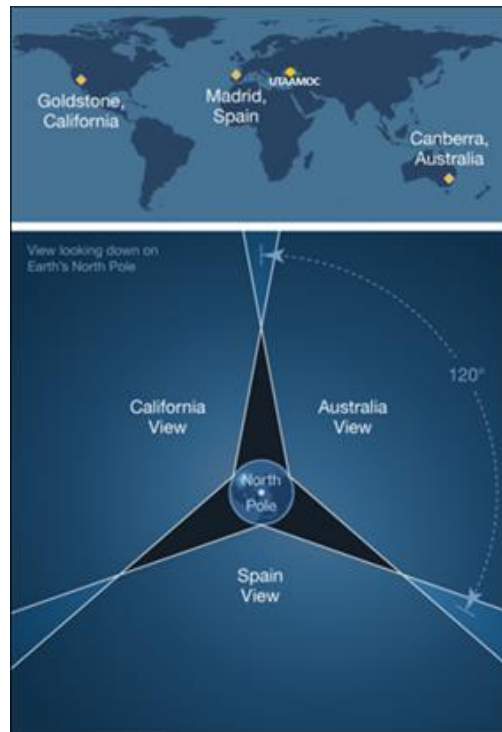


Figure 32:Ground System[D1]

Figure[D1]

The MADEM lander communicates with earth through areostationary relay satellite, and relay satellite communicates with earth via Goldstone, Madrid, UTAA or Canberra ground stations. Mars relay satellite is using two parabolic dishes to communicate with earth and the lander.

Budgets

7.5.1.8 MASS BUDGET

Assign average power consumption to each equipment. Create a power budget based on the figures. You can have multiple budgets for different modes. Calculate solar panel sizes.]

7.5.1.9 POWER BUDGET and Solar Panel Calculations

The power budget of the MADEM mission has two different modes. The first mode is the landing mode, and the second mode is landed mode. To make the power budget calculation, a simulation of the mars' atmosphere, and time frame has used as an input. The solar flux that Mars gets between 6-18 AM has been written to excel and the average of them taken as a solar flux input. The raw power consumption of each equipment taken from the data sheets, and the shifts of the equipment's are given as an average.

Table 16: Power Consumption

Component	#	Power Consumption	24 hours work period (hours)	Day (6-18) (Wh)	Night (18.5-5.5) (Wh)	1 Martian Day (Sol) Wh		
BAE RAD 750 Single-board Computer	1	11-14 W	24	168	168	336		
PANCAM	2	13 W (6.5 W each)	4	26	26	52		
Lower Power Robotic arm	1	10-50 W	2.8	70	70	140		
ChemCam LIBS instrument	1	1-10-50 W	2.4	60	60	120		
Magnetometer	1	10 W	0.033333333	0.166666667	0.166666667	0.333333		
Seismometer	1	2 W	6	6	6	12		
Honeywell S-Band TTC Transceiver STC-MS03	1	18 W	4	36	36	72		
Power Control and Distribution VPCDU-1	1	2.5 W	24	30	30	60		
High Reliable Thermostats	1	5 W	12	0	60	60		
Temperature Sensors 0118MF	1	1 mW	24	0.12	0.12	0.24		
TOTAL Landed Power Consumption				396.2866667	456.2866667	852.5733	-28.8032 Ah	
Gyroscope BNO085	1	0.39 mW	0.011805556			0.001105		
Sun Sensor SITAEL	1	50 mW	0.011805556			0.141667		
Three Axis Accelerometer	1	13.5 W	0.011805556			3.825		
Torque Sensor	1	250 mW	0.011805556			0.708333		
Star Tracker AAC Clyde Space	1	700 mW	0.011805556			1.983333		
S.LDU 345 Panchromatic Camera	1	10 W	0.011805556			2.833333		
TOTAL Landing Power Consumption						477.7328	-16.1396 Ah	
High Power Li-Ion Polymer Battery PN 39381-4	4 (parallel)					2604.8	88 Ah	
Depth of Discharge (DoD) for Sol Night						0.175171	20%	

For the maximum allowable lifespan of the battery and the mission, Depth of Discharge (DoD) of the battery is limited to 20%.

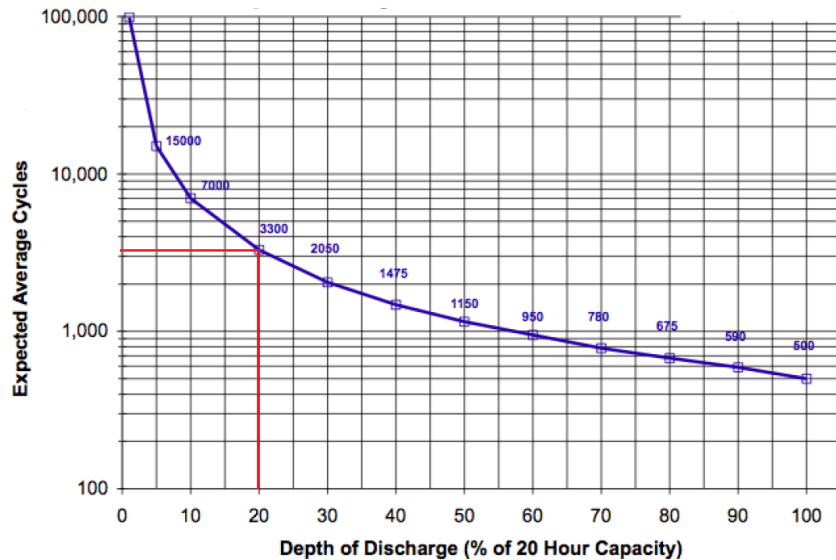


Figure 33: Depth of Discharge

Expected Average Cycles: 3300 Cycles

Expected Mission Lifetime: 2750 Earth Days (7.53 Earth Years)

Solar panel calculations were driven from AST 402 Spacecraft communications, W3_ex1.pdf. codes. To find the total solar cell area and solar panel area, packing factor, solar cell dimensions, and solar efficiency are taken from the datasheet of the solar panel.

```

clc;
clear all;
format short
%Triple Junction Gallium Arsenide Solar Cell efficiency
seff=0.3;
%Packing factor
pf=0.9;
%Solar flux on local slope Ls=290.2 deg (max)
pc=590;
%Total power consumption
pr=1.5e3;
%Cell area
Ac=0.01*0.01;
%Total cell area
Ar=pr/590/seff
%Number of solar cells
nc=Ar/Ac
%Total solar panel area
nc=int32(nc)+1
At=Ar/pf
%Solar panel mass in kg
sp_mass=nc*125*1e-6

```

Total mass of the solar panels:?

Number of solar cells:?

Total area of the solar panels:?

Figure 34:Solar panel calculations codes

7.5.1.10 Thermal Budget

To ensure the functionality of the lander through its mission, a preliminary thermal balance analysis (with comprehensive formula excel table) was created. This thermal balance analysis aimed to predict the temperature distribution across lander during hot, and cold scenarios.

Key parameters were Martian surface conditions (solar irradiance, mars IR emission, mars albedo), material properties (material emissivity, and absorptivity), and heat sources (taken as power dissipation).

Worst-case hot lander temperature

Table 17:Worst-case hot lander temperature

Input	Names	Units	Parameters
Radius of Mars	Rm	km	3396.2
Altitude	H	km	0

View factor, sphere to Mars	Fs-m	-	0.5
View factor, flat plate array to mars	Fa-s	-	1
Factor of Reflection of solar energy off a spherical mars	Ka	-	1
Solar constant	Gs	W/m ²	590
Mars IR emission	qIR	W/m ²	110
Mars albedo	Ma	-	0.25
MADEM electrical power dissipation	Qw	W	396.28
Stefan Boltzmann constant	σ	W/(m ² *K ⁴)	5.67E-08
Emissivity of sphere	ε	-	0.81
Solar absorptivity of sphere	a	-	0.77
Diameter of spherical spacecraft	D	m	2.4
Worst-case hot temperature of MADEM	-	K, °	267.971679
			-5.028 C

Worst-case hot lander temperature: 268 K

Worst-case updated by taking convection from mars atmosphere

Taking average surface pressure as 0.6 kPa == 0.06% Earth' sea level pressure

Heat Loss due to Convection = $0.5 \text{ W}/(\text{m}^2 \cdot \text{K}) * 5 \text{ m}^2 * (268 \text{ K} - 200 \text{ K})$

=34 Watts

= $268 \text{ K} - 34 \text{ W} / (5 \text{ m}^2 * 1 \text{ K/W})$

Updated temperature:

$268 \text{ K} - 34 \text{ W} / (5 \text{ m}^2 * 1 \text{ K/W}) = 234 \text{ K}$

Worst-case cold lander temperature

Table 18:Worst-case cold lander temperature

Input	Names	Units	Parameters
Radius of Mars	Rm	km	3396.2
Altitude	H	km	0
View factor, sphere to Mars	Fs-m	-	0.5
View factor, flat plate array to mars	Fa-s	-	1
Factor of Reflection of solar energy off a spherical mars	Ka	-	1
Solar constant	Gs	W/m ²	0
Mars IR emission	qIR	W/m ²	110
Mars albedo	Ma	-	0.25
MADEM electrical power dissipation	Qw	W	456.286
Stefan Boltzmann constant	σ	W/(m ² *K ⁴)	5.67E-08
Emissivity of sphere	ε	-	0.81

Solar absorptivity of sphere	a	-	0.77
Diameter of spherical spacecraft	D	m	2.4
Worst-case cold temperature of MADEM	-	K, °	197.429846
-75.57 C			

7.5.1.11 Communication Architecture

[Give the communication architecture. Based on data rates for communication, create the link budget and decide on antenna sizes. You can use ready tools for estimating the budget.]

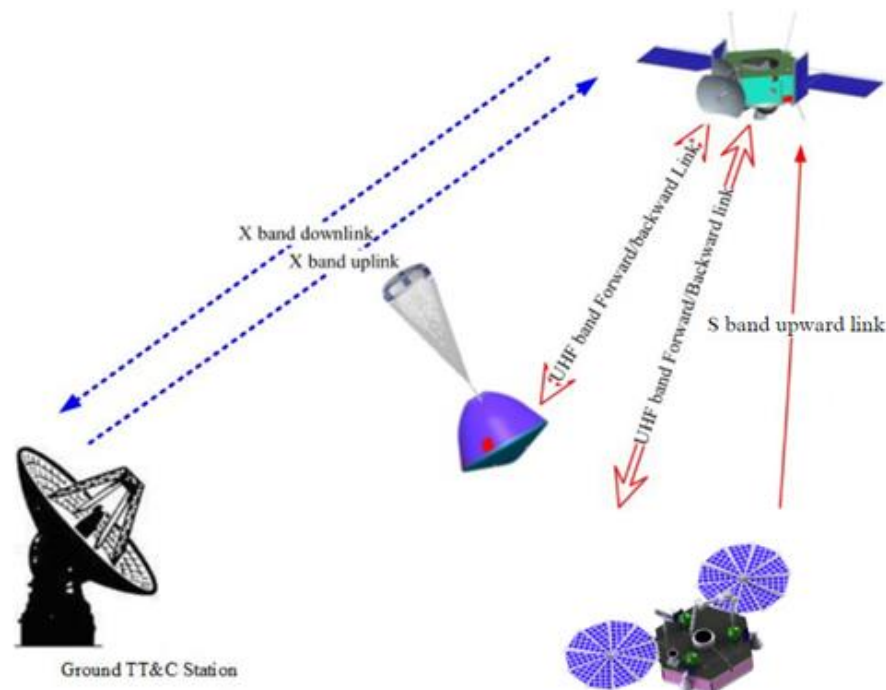


Figure 35:7.5.1.11 Communication Architecture[D2]

In this mission, Uplink and downlink data transfers from/to lander are held by RF links and with mars relay satellite. At landed stage, data transfers of the lander) with relay satellite (interlink) held in S-band. At orbit and entry stages, interlinks are made in UHF band with relay satellite.

7.5.1.12 RF link budget

To ensure enough signal strength is received, a detailed RF link budget analysis was conducted. Parabolic dish equations of the link budget taken from the AST 454 instruction book "Space Mission Analysis and Design". This analysis determined the required antenna gains, and receiver sensitivity to achieve a desired signal-to-noise ratio (SNR) at the receiving relay satellite on the areostationary orbit. The link margin taken as 3 dB. Since antenna types, data rates and data types are different for the mission, two link budget tables are driven.

Table 19:Uplink link budget

MADEM UPLINK LINK BUDGET			
Item	Symbol	Units	Uplink
Bit Error Rate	BER	-	1.00E-05
Frequency	f	Hz	2.25E+09
Data Rate	R	bps	1.23E+07
Transmitter Power	P	dBW	11.46128036
Transmitter Line Loss	Ll	dB	-1
Transmit Antenna Beamwidth	teta	degrees	50
Peak Transmit Antenna Gain	Gpt	dBi	1.05E+01
Transmit Antenna Diameter	D	m	0.190909091
Transmit Antenna Pointing Loss	Lpt	dB	-3
Transmit Antenna Net Gain	Gt	dBi	7.46E+00
Bandwidth		MHz	1.23E+01
Space Loss	Ls	dB	-1.84E+02
Proagation Path Length	S	m	1.70E+07
Transmit Antenna Pointing Offset	et	degrees	25
Equivalent Isotropic Radiated Power	EIRP	dbW	1.79E+01
System Noise Temperature	Ts	K	614
Propagation, polarization loss	La	dB	-0.3
Receiver Gain	Gr	dBi	50
Receiver Diameter	Dr	m	0.5
Eb/No		dB	1.32E+01
Receive Antenna Pointing Loss	Lpr	dB	-0.1
Modulation	-	-	BPSK
Required Eb/No	Eb/No(1)	dB	9.7
Link Margin	-	dB	3.54E+00

Table 20: Downlink link budget

MADEM LINK BUDGET FOR DOWNLINK			
Item	Symbol	Units	Downlink
Bit Error Rate	BER	-	1.00E-08
Frequency	f	Hz	2.03E+09
Data Rate	R	bps	7.00E+01
Transmitter Power	P	dBW	10
Transmitter Line Loss	Ll	dB	-1
Transmit Antenna Beamwidth	teta	degrees	20
Peak Transmit Antenna Gain	Gpt	dBi	1.84E+01
Transmit Antenna Diameter	D	m	0.477272727
Transmit Antenna Pointing Loss	Lpt	dB	-3
Transmit Antenna Net Gain	Gt	dBi	1.54E+01
Bandwidth		MHz	4.00E+00
Space Loss	Ls	dB	-1.83E+02
Proagation Path Length	S	m	1.70E+07
Transmit Antenna Pointing Offset	et	degrees	10
Equivalent Isotropic Radiated Power	EIRP	dbW	2.44E+01
System Noise Temperature	Ts	K	614
Propagation, polarization loss	La	dB	-0.3
Receiver Gain	Gr	dBi	15
Receiver Diameter	Dr	m	0.2
Eb/No		dB	3.81E+01
Receive Antenna Pointing Loss	Lpr	dB	-0.1
Modulation	-	-	BPSK
Required Eb/No	Eb/No(1)	dB	12
Link Margin	-	dB	2.61E+01

7.5.1.13 Data Rate Calculations

Three digital data streams, Science and payload data, engineering data, and command data are calculated to make data rate budget. The data rates of the equipments are taken from the data sheet and combined with the working hours to get the data rates of each. Since data transmission will be from lander to an areostationary relay satellite, there is no need to calculate data transmission rate (since link speed equals to the raw data rate because of the areostationary properties).

The Martian environment can introduce errors into the data streams due to space weather, and weak signal strength for the MADEM. Because of its' error correction capability, high data integrity, and implementation benefits, Reed-Solomon is chosen as a coding scheme.

Table 21:Uplink data rates

UPLINK DATA RATES OF THE MADEM LANDER			
Science or Payload Data	Landing (bps)	Landed (bps)	Worst Case (bps)
Seismometer	0	0.033	12
Magnetometer	0	0.0044	16
Panoramic camera	0	1.30E+07	1.30E+07
Panchromatic camera	3.07E+06	0	4.80E+06
LIBS	0	1.00E+06	1.00E+06
Engineering Data	Landing (bps)	Landed (bps)	Worst Case(bps)
Attitude and Position	480	0.026	480
Temperature	2.13	2.13	64
Health and Status	10.6	2.13	128
Power System	1.36	1.36	6.8
Communications	4	4	4
DATA RATE	2.69517376315789 Mbps	12.2543944591228 Mbps	13.970230826 Mbps (Landed) 3.0726828 Mbps (Landing)

Coding Scheme	Symbol/bit
Reed-Solomon (255, 223)	1.1434978

Table 22:Downlink data rates

DOWNLINK DATA RATE FOR MADEM	
Command Data	bps
Command	70
DATA RATE	62.222222
Coding Scheme	Symbol/bit
BCH (63,56)	1.125

7.6 Assessment of Programmatic Aspects

Overview

The Mars Lander Project is a complex project that involves the design, development, and launch of a lander that will be sent to Mars to collect and return samples of Martian soil and rock. The WBS for this project is a hierarchical structure that breaks down the project into smaller, more manageable tasks.

Levels of the WBS

The WBS for the Mars Lander Project is divided into four main levels:

- **Level 1: Project**
- **Level 2: Major Components**
- **Level 3: Subsystems**

- **Level 4: Work Packages**

Level 1: Project

The project level is the top level of the WBS and represents the overall project. The project is divided into four major components:

- **Safety and Assurance**
- **Payload Development**
- **Lander Integration and Test**
- **System Deployment**

Level 2: Major Components

The major components level represents the four principal areas of the project. Each major component is further divided into subsystems.

- **Safety and Assurance**
 - SSMA Management
 - Quality Assurance
 - Risk Management
- **Payload Development**
 - System Engineering Management
 - Lander Architecture
 - Interface Control
 - Configuration Control
- **Lander Integration and Test**
 - Lander Management
 - Lander Integration and Testing
 - Site Preparation
- **System Deployment**
 - System Engineering
 - Security Validation
 - Operability Implementation

Level 3: Subsystems

The subsystems level represents the smaller components of each major component. Each subsystem is further divided into work packages.

- **SSMA Management**
 - SSMA Plan Development
 - SSMA Implementation
 - SSMA Verification and Validation
- **Quality Assurance**

- Quality Assurance Plan Development
- Quality Assurance Implementation
- Quality Assurance Verification and Validation
- **Risk Management**
 - Risk Management Plan Development
 - Risk Identification and Assessment
 - Risk Mitigation and Monitoring

Level 4: Work Packages

The work packages level represents the smallest units of work in the project. Each work package is a discrete task that can be assigned to a team member or group.

- **SSMA Plan Development**
 - Identify SSMA requirements
 - Develop SSMA plan
 - Review and approve SSMA plan
- **SSMA Implementation**
 - Implement SSMA procedures
 - Conduct SSMA training
 - Monitor SSMA implementation

Symbols and Notations

The WBS for the Mars Lander Project uses symbols and notations to represent different elements of the structure.

- **Boxes** represent tasks or deliverables.
- **Lines** connect tasks or deliverables to show their relationships.
- **Arrows** indicate the direction of the flow of work.
- **Numbers** are used to identify each task or deliverable.

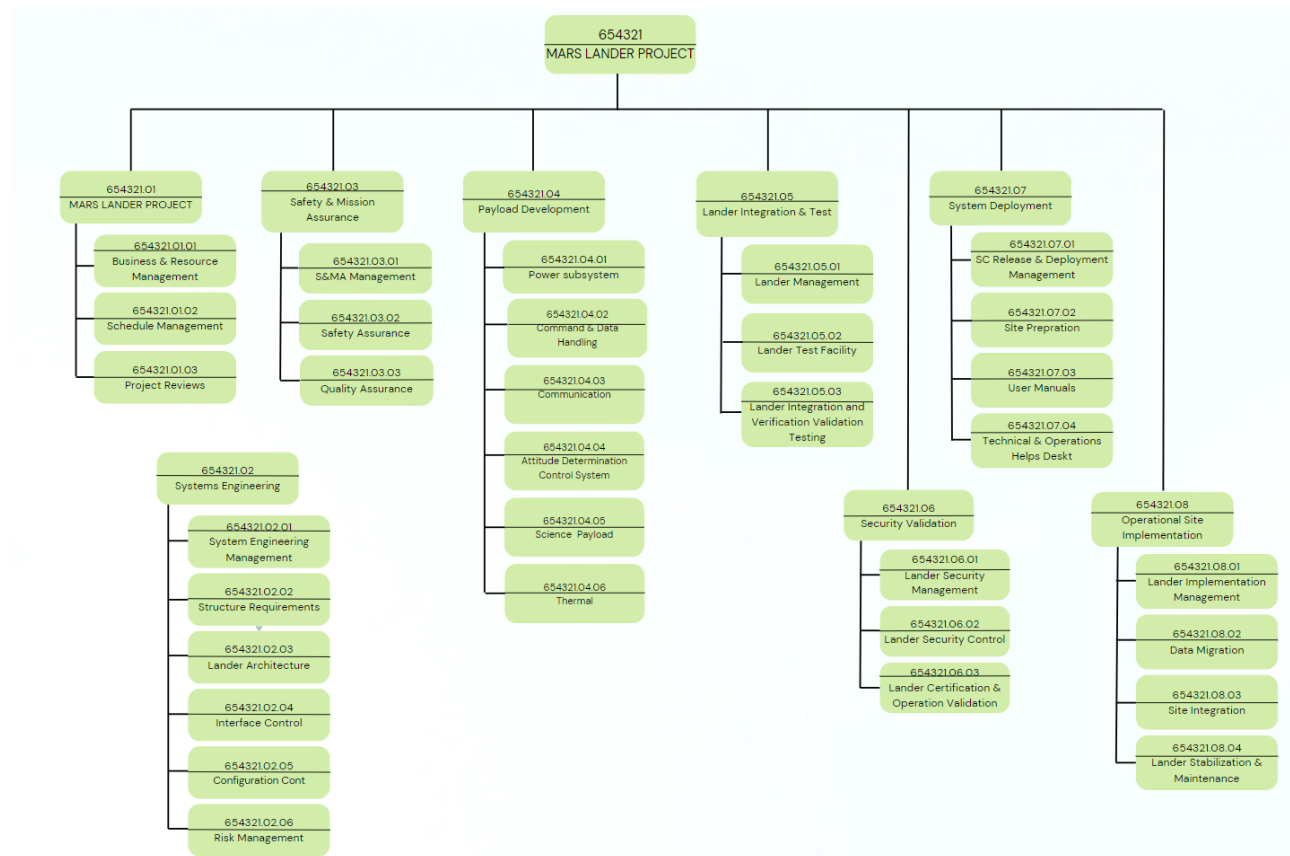
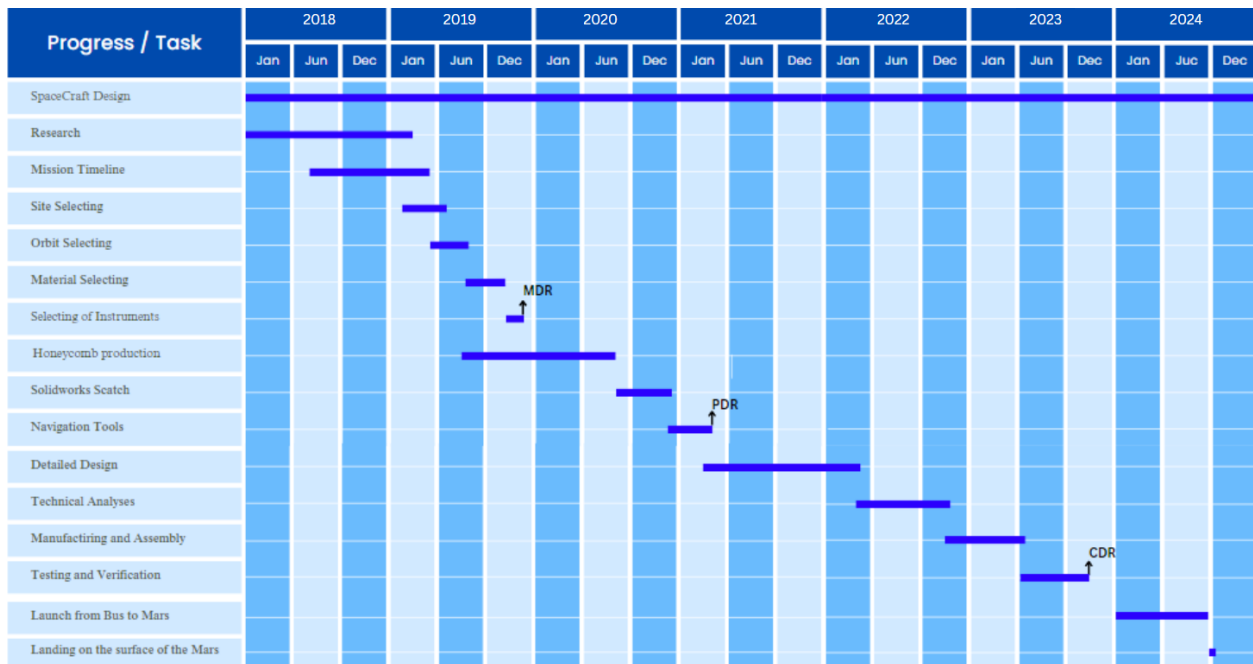


Figure 36: Work Breakdown Structure

7.7 Gantt Chart

Table 23:Gantt Chart



The Gantt chart for the Mars Lander Project is a visual representation of the project schedule. It shows the start and end dates for each task, as well as the dependencies between tasks. The Gantt chart can be used to track the project's progress and identify any potential delays or risks.

Key Elements of the Gantt Chart

The Gantt chart for the Mars Lander Project includes the following key elements:

- **Tasks:** Each task is represented by a horizontal bar on the chart.
- **Start Dates:** The start date for each task is indicated by the left edge of the bar.
- **End Dates:** The end date for each task is indicated by the right edge of the bar.
- **Dependencies:** Dependencies between tasks are indicated by arrows connecting the bars.
- **Milestones:** Milestones are represented by diamonds on the chart.

7.8 Risk Assessment

[risks with likelihood (minimum, low, medium, high, maximum) and severity of impact (negligible, significant, major, critical, catastrophic). You can check [ECSS-M-ST-80C](#) to learn about the process]

Table 24:Risk Assessment

Risk	Failure Mod	Likelihood	Critically
MALFUNCTIONING IN THRUST CONTROL	Dual Failure	HIGH	5
PLASTICALLY DEFORMING DUE TO RETRO-ROCKET SHUT OFF	Multiple Failure	HIGH	4
DEFLECTION OR EVEN TUMBLING CAUSED BY DUST STORMS	Single Failure	MEDIUM	3
CRATER CAUSING	Multiple Failure	LOW	2
A WRONG MEASUREMENT CAUSED BY THE HIGHLY STRESSFUL ENVIRONMENT ENCOUNTERED DURING THE PARACHUTE DEPLOYMENT PHASE, OR SENSITIVITY TO HIGH ANGULAR ACCELERATION	Single Failure	MEDIUM	2
PROPAGATION ERROR FROM GYROS FROM LAST KNOWN ATTITUDE	Single Failure	LOW	3
LARGE CANOPY MOTION DUE TO UNSTEADY WAKE DYNAMICS CAUSING LARGE RISER ANGLE VARIATIONS INCLUDING BRIDLE SLACKING	Single Failure	MEDIUM	1
RISK OF FAILURE TO REACH ORBIT.	Complete Failure	MEDIUM	4
WEATHER CONDITIONS DURING LANDING AND SURFACE OBSTACLES.	Dual Failure	HIGH	5
COMMUNICATION INTERRUPTIONS BETWEEN THE SURFACE VEHICLE AND EARTH.	Multiple Failure	LOW	5
ISSUES WITH SAMPLE COLLECTION AND ANALYSIS EQUIPMENT.	Single Failure	MEDIUM	3
RESISTANCE TO STORMS OR EXTREME TEMPERATURE CHANGES.	Dual Failure	MEDIUM	2

Thermal Control Risk Assessment

Table 25:Thermal Control Risk Assessment

Component	Failure Mode	Effect	Prevention	Criticality
-----------	--------------	--------	------------	-------------

Heat Shield	Structural failure (cracking, burn-through)	Loss of mission, vehicle destruction	Thorough testing, robust design, quality control	5
Multi-layer insulation	Degradation (tearing, punctures)	Reduced thermal performance	Careful handling, inspection, protective layers	3
Paint	Delamination, peeling	Negligible impact on thermal performance	Proper application, adherence testing	1
Heat Pipe	Leakage, blockage	Reduced heat transfer, localized overheating	Robust design, leak testing, redundancy	3
Radiator	Damage (deformation, cracking)	Reduced heat dissipation	Protection from debris, impact testing	3
Thermostat	Failure to regulate temperature	Component overheating/freezing	Backup thermostat, regular calibration testing	2

Communications Risk Assessment

Table 26: Communications Risk Assessment

Risk	Failure Mod	Likelihood	Criticality
Electromagnetic Interference (EMI) - Susceptibility to EMI.	Dual Failure	High	5
Software Error - Error in transceiver software.	Single Failure	Medium	3
Intentional Disruption - Signal jamming or disruption by a malicious actor.	Complete Failure	Low	1
Parabolic Dish Antenna Damage - Partial or complete loss of communication capabilities, potential for degraded data transmission.	Single Failure	Low	2

Communication Link Interference - Disrupted or degraded data transmission, compromised mission objectives, potential safety risks.	Dual Failure	High	5
Transceiver Malfunction - Loss of communication with lander, potential loss of control, compromised mission.	Complete Failure	Medium	4

7.9 Launch

[Choose several launch vehicles. Show volume, weight etc. constraints for the intended orbit for each vehicle. Show vibration levels for each.]

7.10 Verification Program

[Add the verification matrix. Explain your model philosophy. List the main environmental tests you will perform. State the testing facilities that you will use. Show that the chosen facility is appropriate for the test specifications.]

Table 27: Requirement Verification Matrix

Requirement Verification Matrix of MADEM Mars Lander					
MO		Mission Objectives			
	MO-1	MADEM lander shall collect samples of Mars regolith using robotic arm.			
	MO-2	MADEM lander shall take pictures of mars surface using panchromatic camera.			
	MO-3	MADEM lander shall measure seismic activities of Mars using seismometer.			
	MO-4	MADEM lander shall measure magnetic fields of Mars using magnetometer			
	MO-5	MADEM lander shall take videos of Mars in orbit and during landing using RGB video camera.			
MSC		Mission Success Criteria	Verification Method	Status	Planned mass Testing
	MSC-1	MADEM lander shall land on Mars.	Analysis	Designed	N/A
	MSC-2	MADEM lander shall take panchromatic images of Mars.	Testing	Designed	N/A

	MSC-3	MADEM lander shall measure seismic activities and changes in Mars's magnetic field.	Analysis	Designed	N/A
	MSC-4	MADEM lander shall collect and analyze Mars regolith.	Testing	In Progress	N/A
MD		Mission Design	Verification Method	Status	Planned Testing
	MD-1	MADEM lander shall operate for two Earth months.	Analysis	Designed	N/A
	MD-2	An orbiter shall be released at Mars' areostationary orbit and orbiter shall function as a relay to Earth.	Analysis	Designed	N/A
INS		Science Instruments	Verification Method	Status	Planned Testing
	INS-1	The seismometer shall detect seismic activities of Mars up to 1e-9 meters.	Analysis	Designed	N/A
	INS-2	The panchromatic camera shall take medium resolution panchromatic images of Mars surface.	Testing	Designed	N/A
	INS-3	The robotic arm shall collect regolith to analyze thoroughly.	Testing	In Progress	N/A
	INS-4	The magnetometer shall sense changes in the Mars' magnetic field.	Testing	In Progress	N/A
ADC S		Attitude Determination and Control Subsystem	Verification Method	Status	Planned Testing
	ADCS-1	The ADCS shall use a star tracker for fine attitude determination.	Testing	Designed	N/A
	ADCS-2	The ADCS shall utilize a monopropellant thruster system for landing phase.	Testing, Analysis	Designed	N/A
	ADCS-3	The ADCS shall incorporate visual-based navigation for precise positioning.	Testing	Designed	N/A
	ADCS-4	The ADCS shall maintain attitude while entering Mars atmosphere by spin stabilization ballistic control.	Analysis	Designed	N/A
COM		Communications Subsystem	Verification Method	Status	Planned Testing
	COM-1	The COM subsystem shall provide bi-directional communication between Madem lander and relay satellite.	Analysis	Designed	N/A

	COM-2	The COM subsystem shall uplink scientific and engineering data to the relay satellite.	Testing	Designed	N/A
	COM-3	The COM subsystem shall maintain link margin of minimum 3 db.	Analysis	Designed	N/A
	COM-4	The COM subsystem shall downlink command data from the relay satellite.	Testing	Designed	N/A
	COM-5	The COM subsystem shall downlink command data from Earth ground stations only in emergencies.	Testing	Designed	N/A
	COM-6	Communications shall abide by ITU and FCC regulations.	Inspection	In Progress	N/A
CDH		Command and Data Handling Subsystem	Verification Method	Status	Planned Testing
	CDH-1	The CDH Subsystem shall interface with all subsystem components simultaneously.	Analysis	Designed	N/A
	CDH-2	The CDH Subsystem shall be operable during all powered modes of the lander.	Analysis	Designed	N/A
	CDH-3	The CDH subsystem shall have a reliable, radiation-hardened on-board computer.	Testing, Analysis	Designed	N/A
	CDH-4	The CDH subsystem shall receive panchromatic images, RGB images, telemetry and commands.	Testing	Designed	N/A
	CDH-5	The CDH Subsystem shall have a processor watchdog to mitigate Single Event Lockups (SELs) and memory scrubbers to mitigate Single Event Upsets (SEUs).	Testing, Analysis	Designed	N/A
EPS		Electrical Power Subsystem	Verification Method	Status	Planned Testing
	EPS-1	The EPS shall provide for generation, storage, regulation, and control of electrical power to the satellite payloads for the duration of the primary mission.	Inspection	Designed	N/A
	EPS-2	The EPS shall provide an average power of 396.286 W-hrs during sunlight	Testing, Analysis	Designed	N/A
	EPS-3	The EPS shall provide an average power of 516.16 W-hrs during eclipse.	Testing, Analysis	Designed	N/A
	EPS-4	The EPS shall provide loads with regulated power at 28.6 V.	Testing, Analysis	Designed	N/A
	EPS-5	The EPS shall not exceed a DoD of 20% of battery.	Testing, Analysis	Designed	N/A

	EPS-6	The EPS shall provide a continuous, electrically conductive path between each major structural component and the launch vehicle interface to serve as a single-point ground.	Testing	Design ed	N/A
	EPS-7	The EPS shall operate autonomously unless overridden by ground command.	Testing	Design ed	N/A
	EPS-8	The EPS shall use Power Management Software via the battery charge controller to control battery charging and power management modes.	Testing	Design ed	N/A
STR		Structure	Verificati on Method	Status	Planned Testing
	STR-1	Each material on the Materials List shall comply with requirements for outgassing, corrosion resistance, and flammability resistance and buckling.	Inspectio n	Design ed	N/A
	STR-2	The MADEM team shall develop and maintain a Materials List for the preflight hardware that includes both fabricated and vendor-supplied components. The list shall include metallic materials, non-metallic materials, and coatings.	Inspectio n	Verifie d	N/A
	STR-3	The integrated MADEM system shall be designed to withstand the launch vehicle shock and vibroacoustic environment without failure.	Inspectio n	Design ed	N/A
	STR-4	The lander shall be capable of withstanding an acceleration load factor that correspond to worst-case launch load environments, which is a combination of steady state, low-frequency, transient loads, and high-frequency vibration loads	Analysis, Testing	Design ed	N/A
	STR-5	The primary structure of MADEM lander shall be machined and all-metallic.	Inspectio n	Design ed	N/A
	STR-6	Pyrotechnic devices/mechanisms shall be used on the MADEM spacecraft.	Inspectio n	Verifie d	N/A
	STR-7	The MADEM spacecraft shall not use welded joints or cast metallic components	Inspectio n	Design ed	N/A
	STR-8	The MADEM spacecraft shall accommodate the mounting, placement, and structural support needs of all components of engineering subsystems.	Analysis, Testing	Design ed	N/A
	STR-9	Toxic and/or volatile fluids or gasses shall be used	Inspectio n	Verifie d	N/A

TCS		Thermal Control	Verification Method	Status	Planned Testing
	TCS-1	The thermal monitoring system shall monitor all subsystems with thermistors, using redundancy on critical components, and shall trigger heater activation if minimum temperature is reached.	Testing	In Progress	N/A
	TCS-2	Thermal Control Subsystem shall maintain all payload elements and subsystems within their operational temperature limits, as listed in Thermal Detailed Design Document.	Analysis	In Progress	N/A
	TCS-3	The TCS shall be capable of maintaining component and spacecraft thermal requirements within the range of orbits, mission fly and landing considered for the MADEM mission.	Analysis	In Progress	N/A
	TCS-4	The visible camera shall be mounted on a conductive heat sink (ex: aluminum).	Inspection	Designed	N/A
	TCS-5	Heat from the visible camera case to the bracket shall not be blocked by a non-conductive material (ex: plastic)	Inspection	Designed	N/A
	TCS-6	The processor thermal pad must be soldered to an external ground thermal plane.	Testing	Designed	N/A
FSW		Flight Software	Verification Method	Status	Planned Testing
	FSW-1	FSW shall interface between all implementations of control logic for all subsystems.	Analysis	In Progress	N/A
	FSW-2	The FSW shall simultaneously handle all control tasks and procedures without the loss of data.	Testing	Designed	N/A
	FSW-3	FSW shall be flexible for future design changes of other subsystems.	Testing	In Progress	N/A
	FSW-4	FSW shall be capable of uploading and replacing updated versions of control logic of other subsystems.	Testing	In Progress	N/A
	FSW-5	FSW shall be responsible for gathering telemetry from all subsystems.	Testing	In Progress	N/A
	FSW-6	FSW shall handle high-level flight modes.	Testing	In Progress	N/A

	FSW-7	FSW shall respond to ground commands for changing modes.	Analysis	In Progress	N/A
	FSW-8	Flight anomalies shall be detected by error codes for individual subsystems.	Analysis	Designed	N/A
	FSW-9	FSW shall process and respond to uplinked commands from authorized ground stations.			N/A
MOS		Mission Operations System	Verification Method	Status	Planned Testing
	MOS-1	MOS shall be capable of full-duplex communications with MADEM, including all downlinked science data and uplinked commands.	Analysis, Testing	Designed	N/A
	MOS-2	MOS shall provide facilities, hardware, and software for development of testbed and support final integrated test plan	Testing	Designed, In Progress	N/A
	MOS-3	MOS shall be capable of archiving all raw data for duration of primary mission [2 Earth months] and provide support for ground processing and dissemination of science data.	Testing	Designed	N/A
	MOS-4	MOS shall provide Public Outreach in the form of workshops and a website.	Inspection	In Progress	N/A
	MOS-5	MOS shall develop a mission operations system fault tree, and maintain this fault tree throughout operations to minimize single-point failures	Analysis	In Progress	N/A
MC		Mission Control	Verification Method	Status	Planned Testing
	MC-1	MC shall provide command and control for all phases of the MADEM spacecraft development and operations.	Inspection	In Progress	N/A
	MC-2	MC shall provide uplink encoding, downlink decoding, and delivery of data for necessary operational activities.	Testing	In Progress	N/A
	MC-3	MC shall not provide capability to receive and monitor engineering telemetry and science data in real time during testing and flight operations.	Inspection	In Progress	N/A
GDS		Ground Data System	Verification Method	Status	Planned Testing

	GDS-1	The Server shall be the central repository for all data collected by MADEM that is successfully transmitted and received by the ground station.	Inspection	In Progress	N/A
	GDS-2	The Ground Station shall receive mission data and telemetry and post all data to the appropriate workstation computer and server.	Testing	In Progress	N/A
SMP		Science Mission Planning	Verification Method	Status	Planned Testing
	SMP-1	SMP shall provide specified features for imaging to the mission operations team via the mission planning process. Image requests shall be specified via an image request form.	Inspection	Designed	N/A
	SMP-2	SMP shall calculate downlink rate based on available data rate for each intended data collection.	Analysis	Designed	N/A
DA		Data Analysis	Verification Method	Status	Planned Testing
	DA-1	Data Analysis shall be performed on thermal, visible images and seismometer, magnetometer readings stored in the Ground Data System by the science team to yield relevant results.	Analysis	In Progress	N/A
	DA-2	A public website shall be maintained as a database of science mission results for use in the science community, and as a form of public outreach.	Inspection	In Progress	N/A
TFA C		Test Facilities	Verification Method	Status	Planned Testing
	TFAC-1	The MADEM satellite shall be maintained in a Class 100,000 level facility throughout assembly, integration, and test.	Inspection	Verified	N/A
	TFAC-2	All preflight hardware shall be maintained at the Visibly Clean (VC) level.	Inspection	Verified	N/A
	TFAC-3	The clean facility shall be capable of assembling and testing the lander.	Testing	Verified	N/A
	TFAC-4	The lander shall be lifted securely.	Inspection	In Progress	N/A

EGSE		Electrical Ground Support Equipment	Verification Method	Status	Planned Testing
	EGSE-1	EGSE shall perform the following functions: battery charging and discharging while lander is inhibited; inhibit actuation (inhibit/enable satellite); power lander while lander is inhibited; support functional testing of lander, including subsystem level and full "day in the life" testing.	Testing	In Progress	N/A
	EGSE-2	EGSE shall have a main power switch with indicator light.	Inspection	In Progress	N/A
	EGSE-3	All switches and buttons on EGSE shall be clearly labeled, be separated to avoid accidental actuation, and have covers with an automatic-off feature such that when the cover is closed the switch is in the off position.	Inspection	In Progress	N/A
	EGSE-4	Circuit protection (fuses or circuit breakers) shall be installed on primary circuits on the load (not ground/return) lines of the EGSE and accessible for inspection, reset, or replacement. Circuit breaker trips and fuse blows shall be readily detectable by visual inspection and circuit protection shall be clearly marked with voltage present and rated amperage.	Inspection	In Progress	N/A
	EGSE-5	EGSE shall be self-contained and portable.	Inspection, Testing	In Progress	N/A
	EGSE-6	EGSE shall use standard 240 V, 60 Hz, 3 prong "household" power, preferably through a single plug	Inspection, Testing	In Progress	N/A
	EGSE-7	For EGSE batteries, the polarity of battery terminals shall be clearly marked, and ventilation shall be provided to ensure concentrations of vapor do not reach 25% of the lower explosion limit.	Inspection, Analysis, Testing	In Progress	N/A
	EGSE-8	EGSE connectors shall not have exposed pins.	Testing	In Progress	N/A
MGSE		Mechanical Ground Support Equipment	Verification Method	Status	Planned Testing
	MGSE-1	The MADEM team shall provide MGSE for use in integration operations	Inspection	Designed	N/A

	MGSE -2	Tables and workspaces used for integration and testing shall fully secure the lander.	Inspection	In Progress	N/A
	MGSE -3	The tables and workspaces for handling the lander shall be in a class 100,000 clean room environment.	Inspection	Verified	N/A
	MGSE -4	All MGSE shall be designed using a factor of safety of 5.0 for ultimate failure and be proof loaded to twice the design load.	Analysis, Testing	In Progress	N/A
	MGSE -5	The MADEM team shall provide a shipping container that can accommodate the integrated spacecraft and Light band for transport from Türkiye to the launch site.	Inspection, Testing	Designed	N/A
	MGSE -6	The shipping container shall maintain class 100,000 clean conditions.	Inspection	Designed	N/A
	MGSE -7	The shipping container shall measure the shock environment experienced by the lander during shipping using shock sensors in all 3 axes. Approved shock sensor styles are ball and spring or data-logger type shock sensors – sticker-type shock sensors are not allowed.	Inspection, Testing	Designed	N/A
	MGSE -8	The shipping container shall provide shock isolation to the satellite.	Inspection	In Progress	N/A
	MGSE -9	The MADEM team shall provide lifting equipment that can accommodate the integrated MADEM satellite and Light band for movement within the integration facility.	Inspection, Testing	Designed	N/A
	MGSE -10	The lifting harnesses shall be designed to lift the lander from a single point above its center of gravity.	Analysis, Testing	Designed	N/A
	MGSE -11	The lifting equipment shall be constructed out of clean room approved materials and shall not have exposed crevasses or lubricants which could contaminate the clean environment.	Inspection	In Progress	N/A
	MGSE -12	The lifting equipment shall be designed such that it will not contact the Light band during integration and ground handling operations.	Inspection	Designed	N/A

Table 28: Model philosophy

Model Type	Purpose	Functionality	Qualification Testing	Typical Applications (InSight Example)
Engineering Model (EM)	Demonstrate lander operability and functionality on Mars. Verify lander integration with deployment stage and other mission elements. Support ground testing and training for lander operations.	Partially functional with flight-like subsystems (power, communications, thermal control). May use engineering models of scientific instruments.	Limited qualification testing focused on functionality and operability in environment. Thermal-vacuum chamber testing to verify subsystem performance in simulated Martian conditions. Vibration testing to simulate launch vehicle environment.	System integration testing. Training for lander operations team. Verification of lander command and control systems. Early verification of landing sequence procedures and instrument deployment
Qualification Model (QM)	Qualify the lander design for spaceflight and Martian environment. Identify and address potential design flaws before flight model development.	Fully functional with flight-representative subsystems and materials. Engineering models of some scientific instruments may be used.	Extensive qualification testing simulating launch, cruise, and Martian landing conditions. Thermal-vacuum testing at Martian temperature extremes. Vibration testing to simulate launch vehicle environment.	Validate lander can withstand expected loads and stresses of spaceflight. Refine operational procedures based on performance data. Gather data for flight model risk assessment. Qualification of flight-like scientific instruments (if not using engineering models).
Flight Model (FM)	Perform the scientific mission objectives on the Martian surface. Survive and operate for the planned mission duration.	Fully functional with flight-qualified components and scientific instruments.	Most rigorous qualification testing, exceeding expected mission conditions to ensure high margins of safety. All flight-specific components and materials used. Comprehensive testing of all lander subsystems and scientific instruments.	The lander flown to Mars. Performs scientific measurements. Provides platform for scientific instruments

7.11 Reliability

[Assign reliability values to the equipment you will use. If there are no available figures, assign figures using your engineering judgment based on number of times the equipment you use has flown and similar information. In the worst case, assign 0.99 to all equipment. Calculate system reliability at the end of the design life.]

Table 29: Reliability Table -1

Rs = 1- (1-RaRb)(1-Rc) for Non identically, full redundancy					
Rs = Rc [1 - (1 - Ra)(1 - Rb)] for Partial redundancy					
Components	Ra	Rb	Rc	Rd	Reliability

Transmitter	Data Module 0.95	Power Amplifier 0.9	Antenna 0.85	-	0.87225
Receiver	Receiver Module 0.95	Antenna 0.9	De-modulator 0.85	-	0.87225
Panoramic Camera	Camera Module 0.9	Lens 0.85	Shutter 0.92	-	0.9062
Nadir Pointing Panchromatic Camera	Camera Module 0.9	Lens 0.85	Shutter 0.92	-	0.9062
Magnetometer	Magnetometer Sensor 0.94	Data Acquisition System 0.9	Power Supply 0.88	-	0.96272
Seismometer	Seismometer Sensor 0.985	Data Acquisition System 0.97	Power Supply 0.95	-	0.9492725
Robotic Arm with LIBS	Robotic Arm 0.97	LIBS System 0.95	Data Acquisition System 0.9	-	0.89865
Power Control and Distribution Board	Power Supply 0.98	Power Distribution Unit 0.97	Control Unit 0.95	Cooling System 0.96	0.999964
Lithium Ion Battery	Battery Cell 0.99	Battery Management System 0.98	Cooling System 0.96	-	0.99992

Table 30: Reliability Table -2

Solar Array	Solar Panel 0.97	Junction Box 0.98	Inverter 0.95	Monitoring System 0.9	0.999994
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Single Board System	Processor Module 0.98	Memory Module 0.97	Storage Unit 0.95	Power Supply 0.96	0.9999994
Temperature Sensor	Sensor Element 0.9	Signal Conditioning Circuit 0.95	-	-	0.855
Thermostat	Temperature Sensor 0.92	Control Unit 0.96	Relay 0.94	-	0.9558
Gyroscope	Sensing Element 0.999	Control Unit 0.995	Filter Reliability 0.998	-	0.9994995
Accelerometer	Sensing Element 0.998	Control Unit 0.995	Signal Conditioning Circuit 0.995	-	0.998989
Star Tracker	Image Sensor 0.98	Processor Unit 0.97	Star Pattern Recognition Software 0.99	-	0.9698
Sun Sensor	Photodetector 0.98	Signal Conditioning Circuit 0.97	Collimator 0.99	-	0.9698
Torque Sensor	Sensing Element 0.98	Signal Conditioning Circuit 0.97	Calibration Fixture 0.95	-	0.969

8 Conclusion

[“The primary objectives of this review (PDR) are: 1) Verification of the preliminary design of the selected concept and technical solutions against project and system requirements. 2) Release of final management, engineering, and product assurance plans. 3) Release of product tree, work breakdown structure and specification tree. 4) Release of the verification plan (including model philosophy).” Did you realize this task properly? (Note that the second one is excluded.)]

The MADEM lander Preliminary Design Review Report (PDR) has successfully evaluated the chosen concept and its technical solutions for the mission and system requirements. This report focused on verifying feasibility and effectiveness of the design in achieving the design solutions for lander descent, landing, and surface operations, ensuring the MADEM lander can successfully touch down on the Martian surface, the design of the orbiter and its communication capabilities have been evaluated to ensure it functions as a reliable relay satellite between the lander and Earth, while maintaining a minimum link margin of 3 dB for both uplink and downlink. Maintenance of the link margin approved by link budget. The thermal requirements of the MADEM lander to operate in the Martian environment has approved with thermal analysis. The sustainable data rate by given limited parabolic dish antenna dimensions approved by data rate calculations and RF link budget. Reliability of the space system approved by reliability table. The verification matrix of the MADEM lander showed that this system approach is feasible by the tests that done or can done in the given facilities for all requirements.

Furthermore, the PDR has addressed all requirements that given from the user, and from the space agency. This successful PDR paves the way for the release of final management, engineering, and product assurance plans, along with the product tree, work breakdown structure, and verification plan (including model philosophy). These will provide a comprehensive roadmap for the MADEM lander mission, ensuring its successful development and deployment to Mars.

9 References

- [1] J. C. LaCombe, E. L. Wang, M. Nicolescu, P. Rivera and B. Poe, "Design experiences with a student satellite program," in *Proceedings of the 2007 American Society for Engineering Education Pacific Southwest Annual Conference*, Reno, Nevada, USA, 2007.